Error-Bounded Compression For Generative Machine Learning Applications in High-Resolution Numerical Weather Prediction

Semester Project

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Abstract

High-resolution numerical weather prediction and climate simulations generate rapidly increasing data volumes, making storage and input/output costs major constraints for operational and machine-learning workflows. This semester project investigates the Error-Bounded Climate-data Compressor (EBCC) for reducing the storage footprint of MeteoSwiss COSMO-2 analysis data while preserving controlled reconstruction accuracy and downstream usability. EBCC was applied to four variables: 10 m zonal wind, 10 m meridional wind, 2 m temperature, and hourly total precipitation; wind and temperature were compressed directly, while precipitation was compressed after a $\log(1 + x)$ transformation. The implementation integrates EBCC into a Zarr-based workflow by compressing variables independently, processing one two-dimensional time slice per chunk, and parallelizing over time indices with SLURM. All computations were carried out on Balfrin, a high-performance virtual cluster on the Alps infrastructure that is primarily dedicated to critical operational workloads, including MeteoSwiss weather forecasting.

For the selected dataset, the archive size was reduced from 621.640 GiB to 21.485 GiB, corresponding to an aggregate compression ratio of approximately 28.94×. Compression quality was assessed using error statistics, spatial error maps, and an extreme-event case study, showing bounded reconstruction errors concentrated near zero and no evident large-scale spatial bias in the representative diagnostics. Downstream reliability was evaluated with CorrDiff by comparing predictions from original and compressed inputs against COSMO-2 analysis fields. Across selected precipitation, wind, diurnal-cycle, and probability-of-exceedance diagnostics, compressed-input predictions remained close to the original-input baseline. Overall, the results indicate that EBCC can substantially reduce storage requirements for high-resolution weather data while preserving both bounded reconstruction accuracy and the behavior of a representative downstream machine-learning workflow under the tested settings.

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Chapter 1

Introduction

1.1 Motivation: Data Growth in High-Resolution Weather and Climate Simulations

Numerical weather prediction and climate simulations are increasingly limited not only by the cost of computation, but also by the cost of storing, transferring, and repeatedly accessing model output. As spatial and temporal resolutions increase, the number of grid points, prognostic variables, diagnostic variables, ensemble members, and archived time steps grows rapidly. This creates a data-management bottleneck for both operational forecasting centers and research workflows. In particular, high-resolution regional simulations and reanalysis archives can become difficult to use efficiently when downstream tasks require repeated access to large multidimensional arrays.

This issue is especially relevant for machine-learning applications in numerical weather prediction. Modern data-driven models require large training and evaluation datasets, and their performance depends on the availability of physically meaningful fields at sufficient spatial and temporal resolution. However, storing all relevant variables at full precision and full resolution is often inefficient. A compression method is therefore useful only if it reduces data volume while preserving the scientific information required by the intended application. For weather and climate data, this requirement is nontrivial because important information is often contained in extremes, gradients, thresholds, and derived quantities rather than only in average pointwise accuracy.

Lossless compression preserves the original data exactly, but it typically offers limited compression ratios for floating-point geophysical fields. Lossy compression can achieve substantially larger reductions, but uncontrolled loss may distort extremes, introduce spatial artifacts, or affect downstream diagnostics. Error-bounded compression provides a compromise: the decompressed field is allowed to differ from the original field, but the reconstruction error is explicitly constrained. If x denotes the original field and $\hat{x}$ the decompressed field, a maximum absolute error bound can be written as

$$\|x - \hat{x}\|_\infty = \max_i |x_i - \hat{x}_i| \leq \varepsilon, \tag{1.1}$$

where ε is a user-defined tolerance. This form of control is attractive for scientific workflows because the compression error is not left implicit.

The present project studies this idea using the Error-Bounded Climate-data Compressor (EBCC), a method designed specifically for weather and climate applications [1]. EBCC combines a base compression layer with a residual correction layer in order to achieve high compression

ratios while controlling large reconstruction errors. The main motivation of the project is to test whether this approach can be integrated into a MeteoSwiss data workflow and whether the resulting compressed data remain useful not only in direct reconstruction metrics, but also in a downstream machine-learning application.

1.2 Project Objectives and Contributions

The objective of this semester project is to apply EBCC to high-resolution MeteoSwiss COSMO-2 analysis data and evaluate its suitability for machine-learning-oriented numerical weather prediction workflows. The project focuses on four meteorological variables: 10 m zonal wind, 10 m meridional wind, 2 m temperature, and hourly total precipitation. These fields were selected because they are directly relevant to near-surface weather, extreme-event diagnostics, and downstream downscaling applications.

The first objective is to implement a practical compression pipeline around EBCC. The original COSMO-2 analysis data are stored in Zarr format, so the implementation must preserve a Zarr-compatible workflow while allowing EBCC to operate on two-dimensional meteorological fields. Each variable is compressed independently, and each time step is treated as one two-dimensional frame. This leads to a chunking structure of the form

$$(N_t, N_y, N_x) \quad \text{with chunks} \quad (1, N_y, N_x), \tag{1.2}$$

where N_t is the number of time steps and $N_y \times N_x$ is the horizontal grid size. This design matches the frame-wise structure expected by the compressor and supports parallel execution over time indices.

The second objective is to define variable-specific error targets and preprocessing choices. Wind components and temperature are compressed directly, while hourly precipitation is first transformed using

$$y = \log(1 + x), \tag{1.3}$$

where x is the precipitation field. After decompression, the inverse transform

$$\hat{x} = \exp(\hat{y}) - 1 \tag{1.4}$$

is applied. This transformation is used because precipitation is strongly skewed and contains many near-zero values. The error bound for precipitation is therefore imposed in transformed space.

The third objective is to quantify the compression performance. This includes measuring the original and compressed data sizes, computing compression ratios, and inspecting error statistics. The compression ratio is defined as

$$\mathrm{CR} = \frac{S_{\text{original}}}{S_{\text{compressed}}}, \tag{1.5}$$

where S_{original} and $S_{\text{compressed}}$ denote the original and compressed storage sizes, respectively. In addition to aggregate compression ratios, the project evaluates per-variable compression ratios and representative error distributions.

The fourth objective is to assess downstream reliability using CorrDiff, a residual corrective diffusion model for kilometer-scale atmospheric downscaling [2]. The key question is whether CorrDiff predictions generated from compressed inputs remain close to predictions generated from the original inputs. This is tested through spatial maps, precipitation and wind diagnostics, diurnal cycles, and probability-of-exceedance curves. The purpose is not only to determine

whether the compressed data reconstruct the original inputs accurately, but also whether they preserve the behavior of a representative machine-learning workflow.

The main contributions of this project are therefore:

(i) a Zarr-compatible EBCC compression workflow for selected COSMO-2 analysis variables;

(ii) a variable-wise error-target configuration, including log-space compression for precipitation;

(iii) an HPC-parallel compression strategy over independent time indices;

(iv) a quantitative evaluation of compression ratios and reconstruction errors;

(v) a downstream CorrDiff evaluation comparing original-input and compressed-input predictions.

1.3 Scope and Report Structure

The scope of this report is limited to the implementation and evaluation of EBCC for the selected COSMO-2 variables and to the downstream assessment using available CorrDiff diagnostics. The report does not aim to develop a new compression algorithm, modify the internal EBCC method, or retrain CorrDiff. Instead, it focuses on integrating an existing error-bounded compressor into a practical meteorological data workflow and evaluating whether the compressed data remain suitable for representative downstream use.

The report is organized as follows. Chapter 2 introduces the background on weather and climate data compression, error-bounded compression, and the role of CorrDiff in the downstream evaluation. Chapter 3 describes the EBCC method, including its two-layer structure, error-control mechanism, and algorithmic workflow. Chapter 4 presents the COSMO-2 dataset, selected variables, preprocessing choices, Zarr integration, compression script, and HPC execution strategy. Chapter 5 reports compression ratios, error distributions, spatial error maps, and an extreme-event case study. Chapter 6 evaluates downstream CorrDiff predictions using map-based, diurnal-cycle, and distributional diagnostics. Chapter 7 discusses the interpretation of the results, operational advantages, limitations, and recommendations. Chapter 8 concludes the report and outlines possible directions for future work. Additional CorrDiff figures are collected in the appendix to preserve the main report structure while still documenting the full diagnostics.

Chapter 2

Background

2.1 Weather and Climate Data Compression

Weather and climate datasets are high-dimensional by construction. They typically contain several variables over many time steps, vertical levels, horizontal grid points, and, in many cases, ensemble members. As model resolution increases, the storage demand grows rapidly because each refinement of the grid increases the number of spatial degrees of freedom and often motivates more frequent output. This makes storage and input/output performance a central bottleneck in high-resolution numerical weather prediction and climate workflows.

Compression addresses this bottleneck by reducing the amount of data that must be stored and transferred. In weather and climate applications, however, compression cannot be evaluated only by the final file size. The compressed data must remain useful for scientific analysis, operational diagnostics, and downstream models. A method that gives a high compression ratio but distorts extremes, gradients, or threshold-based indices may be unsuitable even if its average reconstruction error is small.

This is particularly relevant for near-surface weather variables. Wind components, temperature, and precipitation are often used directly in applications, but they are also used to derive further diagnostics such as wind speed, wet-hour frequency, high-percentile precipitation, and exceedance probabilities. The quality of a compressed dataset must therefore be judged not only by pointwise reconstruction error, but also by the behavior of such derived quantities.

2.2 Lossless, Lossy, and Error-Bounded Compression

Lossless compression reconstructs the original data exactly. If x denotes the original field and $\hat{x}$ the decompressed field, then lossless compression satisfies $\hat{x} = x$ at every grid point. This property is valuable when bitwise reproducibility is required, but it usually limits the achievable compression ratio for floating-point meteorological fields.

Lossy compression allows the decompressed field to differ from the original field. This can produce much larger storage reductions, but it also introduces the question of how much error is acceptable. Common measures include the root mean square error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2}, \tag{2.1}$$

and the maximum absolute error,

$$\|x - \hat{x}\|_\infty = \max_i |\hat{x}_i - x_i|. \tag{2.2}$$

The RMSE describes the typical size of the reconstruction error, while the maximum absolute error controls the largest pointwise deviation.

Error-bounded compression is a form of lossy compression in which a prescribed error tolerance is enforced. In this project, the relevant bound is an absolute maximum-error target,

$$\|x - \hat{x}\|_\infty \leq \varepsilon, \tag{2.3}$$

where ε is chosen separately for each variable. This form of control is useful because it gives the user an explicit guarantee on the local reconstruction error, rather than relying only on global average metrics.

The corresponding storage reduction is reported through the compression ratio,

$$\mathrm{CR} = \frac{S_{\text{original}}}{S_{\text{compressed}}}, \tag{2.4}$$

where S_{original} and $S_{\text{compressed}}$ are the original and compressed storage sizes. A useful compression method should provide a high compression ratio while keeping reconstruction errors within scientifically acceptable limits.

2.3 Scientific Requirements for Compressing Meteorological Fields

Meteorological compression is variable-dependent. Temperature, wind, and precipitation have different units, numerical ranges, distributions, and scientific roles. A uniform tolerance across all variables is therefore rarely meaningful. For example, an absolute error of a few units may be acceptable for some wind or temperature applications, but precipitation requires more careful treatment because its distribution is sparse, non-negative, and strongly skewed.

The scientific requirements also depend on the diagnostics of interest. Some diagnostics, such as averages, may be relatively insensitive to small pointwise perturbations. Others are more sensitive. Threshold-based quantities can change when values close to the threshold are perturbed, and high-percentile diagnostics can be affected by comparatively few grid points in the distributional tail. This is one reason why precipitation extremes and probability-of-exceedance curves are emphasized later in the report.

Spatial structure is another requirement. Compression errors should not introduce artificial block patterns, large-scale bias, or coherent regional artifacts. For this reason, the evaluation in this project includes both numerical error statistics and spatial error maps. The same principle motivates the downstream evaluation: a compressed dataset is useful only if it preserves the behavior of the workflows that will consume it.

2.4 EBCC in the Context of Existing Compression Methods

The Error-Bounded Climate-data Compressor (EBCC) is a compression method developed for weather and climate applications [1]. Its design addresses a practical limitation of many lossy

methods: even when average error metrics are good, rare large errors may remain problematic for extremes or derived variables. EBCC therefore combines high compression with explicit control of large reconstruction errors.

The method uses a two-layer strategy. A base compression layer captures most of the field at a high compression ratio. A residual layer then corrects the remaining errors that would otherwise exceed the prescribed tolerance. In simplified terms, the decompressed field is obtained by combining a base reconstruction with a residual correction,

$$\hat{x} = x_{\text{base}} + \hat{r}. \tag{2.5}$$

The base layer is responsible for most of the data reduction, while the residual layer targets the long-tail errors left by the base reconstruction.

In this project, EBCC is not modified at the algorithmic level. Instead, it is used as the core compression method in a MeteoSwiss workflow. The project-specific contribution is the integration of EBCC with a Zarr-based COSMO-2 dataset, the selection of variable-wise tolerances, the preprocessing of precipitation, and the evaluation of both reconstruction-level and downstream machine-learning effects. The reference implementation is provided by SPCL [3], and the project implementation is organized in the semester-project repository [4].

2.5 Machine-Learning Workflows for Numerical Weather Prediction

Machine-learning workflows in numerical weather prediction increasingly rely on large structured datasets. These workflows may involve training, validation, inference, downscaling, post-processing, and diagnostic evaluation. Their data requirements can be substantial because they often require repeated access to long time periods and multiple variables.

Compression can support such workflows by reducing storage and input/output costs. However, the relevant question is not only whether the decompressed input fields are close to the original fields. A downstream machine-learning model may respond to small input perturbations in ways that are not obvious from the input error alone. Depending on the model and diagnostic, perturbations may be damped, amplified, or transformed into different spatial patterns.

For this reason, this project evaluates EBCC in two stages. First, the compressed COSMO-2 fields are assessed directly through compression ratios, error distributions, and spatial error maps. Second, the compressed fields are used as inputs to a downstream machine-learning workflow, and the resulting predictions are compared with predictions generated from the original data. This two-stage evaluation is intended to distinguish compression fidelity from application-level reliability.

2.6 CorrDiff as a Downstream Evaluation Model

CorrDiff is used in this project as the downstream machine-learning evaluation model. It is a residual corrective diffusion machine learning model for kilometer-scale atmospheric downscaling [2]. Conceptually, the model combines a regression component, which represents the large-scale conditional mean, with a diffusion-based correction, which restores fine-scale structure and variability.

The comparison in this report is organized around three outputs: the COSMO-2 analysis target, the CorrDiff prediction obtained from original inputs, and the CorrDiff prediction obtained

from EBCC-compressed inputs. The purpose is to assess whether replacing original inputs by compressed inputs materially changes the resulting high-resolution diagnostics.

The evaluation focuses on quantities that are relevant to weather applications and potentially sensitive to compression effects. These include precipitation percentiles, wet-hour frequency, wind-speed diagnostics, diurnal cycles, and probability-of-exceedance curves. If the compressed-input CorrDiff predictions remain close to the original-input predictions across these diagnostics, this provides evidence that the compressed archive preserves the information needed by the tested downstream workflow.

Thus, CorrDiff serves as an application-level stress test for EBCC-compressed data. It links the compression problem to the central practical question of the project: whether strongly compressed, error-bounded meteorological fields can still support machine-learning-based high-resolution weather diagnostics.

Chapter 3

The EBCC Method

3.1 Overview of Error-Bounded Climate-Data Compression

The Error-Bounded Climate-data Compressor (EBCC) is a lossy compression method designed specifically for weather and climate data. Its purpose is to achieve large storage reductions while keeping the reconstruction error under an explicit user-defined tolerance. The method is motivated by the observation that weather and climate datasets are not only large, but also scientifically sensitive: errors in rare events, sharp gradients, or derived diagnostics can be more important than the average pointwise error alone.

EBCC follows a two-layer design. The first layer performs most of the data reduction using a transform-based base compressor. The second layer encodes a residual correction so that large errors left by the base layer are reduced and the requested error bound is satisfied. This structure distinguishes EBCC from a purely rate-controlled lossy compressor: the final objective is not only a high compression ratio, but a high compression ratio under an explicit error constraint.

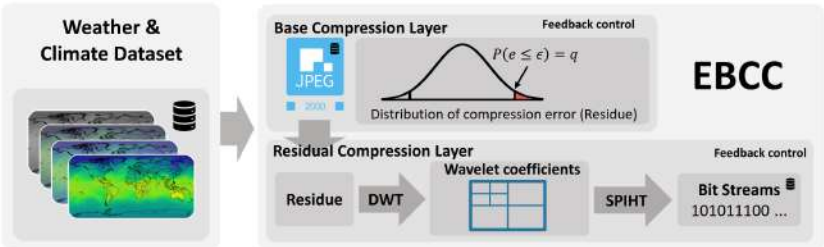


Figure 3.1: Schematic overview of the EBCC method. The compressor uses a base compression layer for the bulk of the data and a residual compression layer to correct excessive errors. Both layers are controlled by feedback mechanisms to satisfy the target error bound. Figure adapted from Huang et al. [1].

The schematic in Figure 3.1 summarizes the main idea. A weather or climate field is first compressed by the base layer. The resulting reconstruction is compared with the original field, and the remaining error is passed to the residual layer. The residual layer is designed to correct the long-tail part of the error distribution, namely the rare but potentially important errors that exceed the prescribed tolerance.

3.2 Two-Layer Compression Strategy

3.2.1 Base Layer: JPEG2000 Transform Coding

The base layer of EBCC uses JPEG2000-style transform coding. This layer is responsible for compressing the majority of the signal. In weather and climate fields, large-scale spatial structures usually contain much of the field energy, and transform coding can represent these structures efficiently. The base layer therefore provides most of the compression ratio.

However, the base layer alone is not sufficient for strict error-bounded reconstruction. Transform-based lossy compression can yield good average accuracy, for example low RMSE, while still producing rare large pointwise errors. These errors are especially relevant in meteorological applications because extremes, gradients, and threshold-based indices may depend on relatively few grid points. EBCC therefore treats the base layer as the first approximation rather than the final compressed representation.

If x denotes the original field and x_b the base-layer reconstruction, the base-layer error is

$$e_b = x - x_b. \tag{3.1}$$

The role of the residual layer is then to encode the information needed to reduce this error so that the final reconstruction satisfies the prescribed bound.

3.2.2 Residual Layer: Wavelet Transform and SPIHT Encoding

The residual layer targets the portion of the base-layer error that remains problematic after the first compression stage. In EBCC, this residual is transformed using a discrete wavelet transform and encoded using SPIHT, the Set Partitioning in Hierarchical Trees algorithm. The purpose of this stage is to efficiently represent the sparse correction associated with large residual errors.

The final reconstruction may be viewed as

$$\hat{x} = x_b + \hat{r}, \tag{3.2}$$

where x_b is the decompressed base-layer field and $\hat{r}$ is the decoded residual correction. The residual layer is therefore not intended to re-encode the entire original field. Instead, it corrects the error left by the base compressor, with particular emphasis on the long-tail errors that would violate the target tolerance.

This design is important for weather and climate data. If a compressor distributes errors broadly or leaves isolated large deviations, derived fields may be affected even when the mean error is small. By explicitly encoding residual corrections, EBCC aims to retain high compression while avoiding excessive outliers in the reconstruction error.

3.2.3 Feedback-Based Rate Control

A central part of EBCC is feedback-based rate control. The method does not assume that a desired error tolerance can be translated directly into a fixed compression parameter in advance. Instead, it adjusts the compression settings iteratively by checking the reconstruction error against the target.

In the base layer, feedback control is used to select a compression ratio that leaves only a controlled fraction of errors above the desired tolerance. In the residual layer, the encoded SPIHT bitstream is truncated at a point that is sufficient to satisfy the final error bound. This procedure allows EBCC to adapt to the structure of the input field rather than applying a fixed compression setting uniformly.

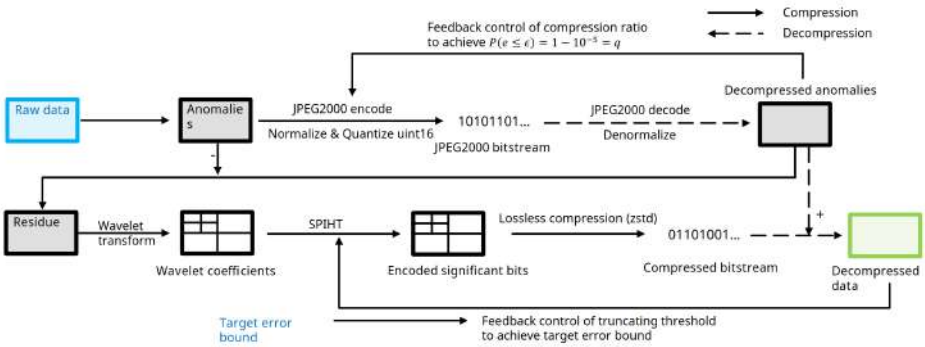


Figure 3.2: Algorithmic workflow of EBCC compression, including base compression, residual formation, wavelet transformation, SPIHT encoding, and feedback-based control of the final reconstruction error. Figure adapted from Huang et al. [5].

The workflow shown in Figure 3.2 highlights that EBCC is not a single-pass fixed-rate compressor. It is instead an error-controlled procedure in which compression parameters are adjusted until the decompressed field satisfies the prescribed tolerance.

3.3 Error Bounds and Rate-Distortion Trade-Off

3.3.1 Maximum Absolute Error Bounds

The main error-control mode used in this project is the maximum absolute error bound. For an original field x and reconstructed field $\hat{x}$, the bound is written as

$$\max_i |x_i - \hat{x}_i| \leq \varepsilon, \tag{3.3}$$

where ε is the selected tolerance. This tolerance is set separately for each variable because meteorological variables differ in units, distributions, and scientific interpretation.

The usefulness of Equation (3.3) is its direct local interpretation. It prevents any single reconstructed grid point from deviating by more than the chosen tolerance in the compressed representation space. In this project, wind components and temperature are bounded in their physical units, while precipitation is bounded after the $\log(1 + x)$ transform.

3.3.2 Long-Tail Error Suppression

A central motivation for EBCC is that average error metrics alone do not fully characterize compression quality. A compressor may achieve a low RMSE while still producing rare large errors. These rare errors form the tail of the error distribution and may matter disproportionately for weather and climate applications.

EBCC addresses this issue through the residual compression layer. The base layer is allowed to perform aggressive compression, while the residual layer corrects excessive errors. In practice, this is intended to produce an error distribution that is concentrated near zero and constrained by the selected maximum error bound. This behavior is particularly relevant when evaluating high-percentile precipitation, wind maxima, and other tail-sensitive diagnostics.

3.3.3 Implications for Extreme Weather Diagnostics

Extreme weather diagnostics often depend on the tails of the distribution. Examples include high precipitation percentiles, maximum wind speed, threshold exceedance probabilities, and accumulated precipitation indices. These quantities may be sensitive to relatively small numbers of grid points, especially when the diagnostic focuses on rare events.

For this reason, an error-bounded compressor is attractive in this setting. It does not guarantee that all downstream diagnostics are unchanged, since nonlinear diagnostics can still respond to bounded perturbations. However, it does ensure that the input perturbation is controlled pointwise. The downstream evaluation in later chapters is therefore essential: it tests whether EBCC-compressed inputs preserve the behavior of CorrDiff predictions for the diagnostics of interest.

3.4 EBCC Algorithmic Workflow

3.4.1 Chunk Preparation

EBCC operates on chunks of multidimensional weather or climate data. In the original EBCC formulation, multidimensional chunks can be reshaped into two-dimensional arrays before compression. In this project, the selected COSMO-2 fields are already processed as two-dimensional horizontal frames, one variable and one time step at a time. This matches the compressor naturally.

Each compressed object therefore corresponds to a field of shape

$$(N_y, N_x),$$

where N_y and N_x are the horizontal grid dimensions. In the Zarr archive, this corresponds to chunking each variable array as

$$(1, N_y, N_x),$$

so that each time index is compressed independently. This design supports both variable-wise error control and embarrassingly parallel execution over time.

3.4.2 Base Compression

After chunk preparation, the base layer compresses the input frame using JPEG2000-style transform coding. This produces a compressed byte stream and a preliminary reconstruction. The reconstruction is compared with the original field to estimate the base-layer error.

The purpose of this step is to remove as much data volume as possible while retaining the large-scale structure of the field. The base layer is therefore responsible for most of the achieved compression ratio. However, since it may leave large pointwise residuals, its output is passed to the residual correction stage.

3.4.3 Residual Encoding

The residual is computed as the difference between the original field and the decompressed base-layer field. This residual is then transformed into wavelet coefficients and encoded using SPIHT. The SPIHT bitstream can be truncated, which allows the method to control how much residual information is retained.

The residual stage is guided by the final error target. The compressor searches for the smallest residual encoding sufficient to make the reconstructed field satisfy the maximum error bound. This is the stage that turns a high-ratio base compressor into an error-bounded compressor.

3.4.4 Decompression and Reconstruction

During decompression, the base layer is decoded first. The residual bitstream is then decoded, transformed back to physical space, and added to the base reconstruction. The final output is

$$\hat{x} = x_b + \hat{r}.$$

The result is a decompressed field that should satisfy the prescribed maximum error bound, subject to the compression configuration and the space in which the bound was imposed.

In this project, this reconstruction is subsequently interpreted according to the variable being processed. For wind and temperature, the decompressed field is used directly. For precipitation, the decompressed field is first interpreted in log-space and then transformed back using the inverse $\log(1 + x)$ transformation. This distinction is important when interpreting the precipitation error tolerance and its downstream effects.

Chapter 4

Dataset and Computational Workflow

4.1 COSMO-2 Dataset and Selected Variables

The dataset used in this project is based on MeteoSwiss COSMO-2 analysis data stored in Zarr format. The compression workflow was applied to a subset of near-surface meteorological variables that are directly relevant for both weather diagnostics and downstream machine-learning applications. The selected variables are listed in Tab. 4.1.

Variable in report	Variable in code	Physical meaning	Unit
`U_10M`	`10u`	10 m zonal wind component	$\mathrm{m\,s^{-1}}$
`V_10M`	`10v`	10 m meridional wind component	$\mathrm{m\,s^{-1}}$
`T_2M`	`2t`	2 m air temperature	K
`TOT_PREC_1H`	`tp`	Hourly total precipitation	mm

Table 4.1: Selected COSMO-2 variables used in the EBCC compression workflow. The report names follow meteorological notation, while the code names correspond to the variable labels used in the source Zarr archive.

The selected dataset covers the time window from `2019-07-03T06:00:00` to `2024-09-04T23:00:00`, corresponding to 45 365 measurements per variable in the experiment reported later. Each variable is stored as a sequence of two-dimensional horizontal fields. The compression workflow treats each time step and each variable independently, so that one compressed chunk corresponds to one meteorological field at one time index.

This variable-wise and frame-wise organization was chosen for three reasons. First, it avoids mixing variables with different units and physical ranges. Second, it allows different error tolerances to be assigned to different variables. Third, it gives a natural parallelization strategy, since time indices can be compressed independently.

4.2 Error Targets and Variable Preprocessing

EBCC was configured using variable-specific absolute error targets. Wind components and temperature were compressed directly, while precipitation was transformed before compression. The targets used in the implementation are listed in Tab. 4.2.

Variable	Preprocessing	Error control type	Target
`10u`	None	Absolute	2.0
`10v`	None	Absolute	2.0
`2t`	None	Absolute	2.0
`tp`	$\log(1+x)$	Absolute in log-space	5×10^{-4}

Table 4.2: Variable-specific EBCC error targets used in the project implementation. The precipitation tolerance is imposed after applying the $\log(1+x)$ transformation.

The precipitation transform is used because hourly precipitation is non-negative, sparse, and strongly skewed. Directly compressing such a field can be inefficient because most entries are near zero while a small number of grid points contain larger values. The encoding function used for precipitation is

$$y = \log(1+x), \tag{4.1}$$

where x is the original precipitation field. After decompression, the inverse transformation is applied:

$$\hat{x} = \exp(\hat{y}) - 1. \tag{4.2}$$

The error bound for precipitation is therefore not imposed directly on x, but on the transformed variable y. This distinction is important when interpreting precipitation errors and downstream precipitation diagnostics.

For wind and temperature, no preprocessing was applied. The decompressed values are therefore interpreted in the same physical units as the original fields.

4.3 Zarr Integration and EBCC Filter Configuration

The source data are stored in a Zarr archive, while EBCC is provided as a filter-oriented compression implementation. The project workflow bridges this difference by wrapping the EBCC filter in a Zarr-compatible codec. Each output variable is written as a separate Zarr array under the destination root directory, with shape

$$(T, N_y, N_x),$$

where T is the number of time steps and $N_y \times N_x$ is the horizontal grid size.

The chunk layout is

$$(1, N_y, N_x),$$

so each chunk contains one complete two-dimensional field. This is the natural layout for the present EBCC workflow because the compressor receives one full horizontal frame at a time. It also keeps the compressed representation aligned with the temporal parallelization strategy used in the HPC workflow.

The EBCC filter is configured independently for each variable. In the implementation, the base compression ratio is set as

```
BASE_CR = 5
```

and the residual optimization mode is set to a maximum error target:

```
RESIDUAL_OPT_NAME = "max_error_target"
```

For each variable, the corresponding tolerance is passed to the EBCC filter through the residual configuration. This gives variable-wise control of the final reconstruction error.

4.4 Compression Script and Codec Wrapper

The main compression script processes one time index per invocation. It opens the source Zarr archive, reads the selected variables, reshapes each variable into a frame-sized chunk, applies variable-specific preprocessing, and writes the result to the destination Zarr archive using the EBCC codec wrapper.

The central preprocessing functions are:

```
def encode_var(var_name: str, arr: np.ndarray) -> np.ndarray:
    if var_name == "tp":
        # Assumes precipitation is non-negative
        return np.log1p(arr)
    return arr


def decode_var(var_name: str, arr: np.ndarray) -> np.ndarray:
    if var_name == "tp":
        return np.expm1(arr)
    return arr
```

Listing 4.1: Variable-specific encoding and decoding functions used in the compression script.

The variable-wise error targets are stored in a dictionary:

```
BASE_CR = 5
RESIDUAL_OPT_NAME = "max_error_target"
VAR_ATOL = {
    "10u": 2.0,
    "10v": 2.0,
    "2t": 2.0,
    "tp": 0.0005,
}
```

Listing 4.2: Variable-specific absolute error targets used in the EBCC configuration.

For each variable, the EBCC filter is initialized with the horizontal field dimensions and the selected error target:

```
ebcc_filter = EBCC_Filter(
    base_cr=BASE_CR,
    height=NY,
    width=NX,
    residual_opt=(RESIDUAL_OPT_NAME, VAR_ATOL[var_name]),
)
zarr_filter = EBCCZarrFilter(ebcc_filter.hdf_filter_opts)
```

Listing 4.3: EBCC filter initialization for one selected variable.

The destination Zarr array is opened with full-frame chunks and the wrapped EBCC codec:

```
var_zarray = zarr.open_array(
    args.dst_path / var_name,
    shape=(T, NY, NX),
    chunks=(1, NY, NX),
    dtype=data.dtype,
    codecs=[AnyNumcodecsArrayBytesCodec(zarr_filter)],
    mode="a"
```

```
)
```

Listing 4.4: Creation of the destination Zarr array with frame-sized chunks and EBCC codec wrapper.

The field at the selected time index is then extracted, encoded if needed, and written to the output array:

```
var_data = data[args.time_index][variables.index(var_name), :, :].reshape((1, NY, NX))
var_data = encode_var(var_name, var_data)
var_zarray[args.time_index:args.time_index+1] = var_data
```

Listing 4.5: Extraction and compression of one variable at one time index.

In the appendix, the full script is included as a complete listing. The main text keeps only the essential parts needed to explain the design of the workflow.

4.5 HPC Parallelization Workflow

The compression task is embarrassingly parallel over time indices because each two-dimensional field is compressed independently. The SLURM workflow exploits this by assigning different time indices to different tasks. Each task processes one timestamp and compresses all selected variables for that timestamp.

The relevant structure of the batch script is:

```
ntasks=${SLURM_NTASKS}
total_time_indices=$((T_STOP - T_START + 1))

offset=0
while [ "${offset}" -lt "${total_time_indices}" ]; do
    global_offset=$((T_START + offset))

    srun -u bash -c "
        time_index=\$(( ${global_offset} + SLURM_PROCID ))
        if [ \"\$time_index\" -le \"${T_STOP}\" ]; then
            python main.py ${DST_ROOT} --variables 2t,10u,10v,tp --time_index \
                $time_index
        fi
    "

    offset=$((offset + ntasks))
done
```

Listing 4.6: Structure of the SLURM compression workflow over time indices.

This loop advances through the full time range in blocks of size equal to the number of SLURM tasks. Within each block, task p processes the time index

$$t = t_{\text{start}} + \text{offset} + p.$$

Since no task depends on the result of another task, this strategy scales naturally until file-system contention or I/O throughput becomes limiting.

In the reported workflow, the batch script uses the `postproc` partition, 128 tasks, and two CPUs per task. The destination archive is written incrementally, with each task filling the relevant time slice for each selected variable.

4.6 Implementation Limitations

The implementation was designed as a practical project pipeline rather than a general-purpose compression framework. Several limitations should therefore be noted.

First, the tolerance values were selected empirically. They are suitable for the tested setting, but they should not be interpreted as universally optimal values for all applications. Different downstream diagnostics, variables, or time periods may require different tolerances.

Second, the precipitation error target is imposed in log-space. This improves the handling of sparse and skewed precipitation fields, but it also means that the tolerance is not a direct absolute precipitation error in physical space. The downstream precipitation diagnostics must therefore be interpreted with this transformation in mind.

Third, the workflow compresses variables independently. This is consistent with the variable-wise error-control strategy, but it does not explicitly preserve cross-variable relationships during compression. Such relationships are instead assessed indirectly through the downstream CorrDiff diagnostics.

Fourth, the current implementation processes one full horizontal frame per chunk. This is appropriate for the present dataset and workflow, but other datasets may require different chunking strategies to balance compression efficiency, random access, and I/O performance.

Finally, the implementation focuses on compression and downstream evaluation rather than on a detailed performance optimization of the compressor itself. The parallelization is effective because the workload is independent over time, but the achieved throughput can still be limited by filesystem behavior, metadata operations, and concurrent writes to the destination archive.

Chapter 5

Compression Performance and Reconstruction Accuracy

5.1 Compression Ratios

The first level of evaluation concerns the storage reduction obtained by applying EBCC to the selected COSMO-2 variables. The compressed time window extends from `2019-07-03T06:00:00` to `2024-09-04T23:00:00`, corresponding to 45 365 measurements per variable. Four variables were compressed: `U_10M`, `V_10M`, `T_2M`, and `TOT_PREC_1H`. Each variable originally occupied 155.410 GiB, giving a total original size of 621.640 GiB.

After compression, the total size was reduced to 21.485 GiB. This corresponds to an aggregate compression ratio of approximately 28.94×, or equivalently a storage reduction of about 96.54%. The compression ratio is computed as

$$\mathrm{CR} = \frac{S_{\text{original}}}{S_{\text{compressed}}}, \tag{5.1}$$

where S_{original} and $S_{\text{compressed}}$ are the original and compressed storage sizes, respectively.

Dataset / Variable	Original Size (GiB)	Compressed Size (GiB)	Compression Ratio
Total	**621.640**	**21.485**	28.9400×
`U_10M`	155.410	6.791	22.8851×
`V_10M`	155.410	6.394	24.3074×
`T_2M`	155.410	6.626	23.4560×
`TOT_PREC_1H`	155.410	1.674	92.8419×

Table 5.1: Original and compressed storage sizes for the selected COSMO-2 variables. The total compression ratio is approximately 28.94×, corresponding to a storage reduction of about 96.54%.

The wind and temperature variables obtain similar compression ratios, ranging from approximately 22.9× to 24.3×. The precipitation field achieves a substantially larger compression ratio of approximately 92.8×. This is consistent with the structure of hourly precipitation fields, which are sparse and dominated by near-zero values. The $\log(1 + x)$ preprocessing further supports efficient representation by reducing the dynamic range of large precipitation values.

These results show that the selected EBCC configuration substantially reduces the storage requirement while retaining explicit error control. The remaining sections examine whether the corresponding reconstruction errors are well behaved.

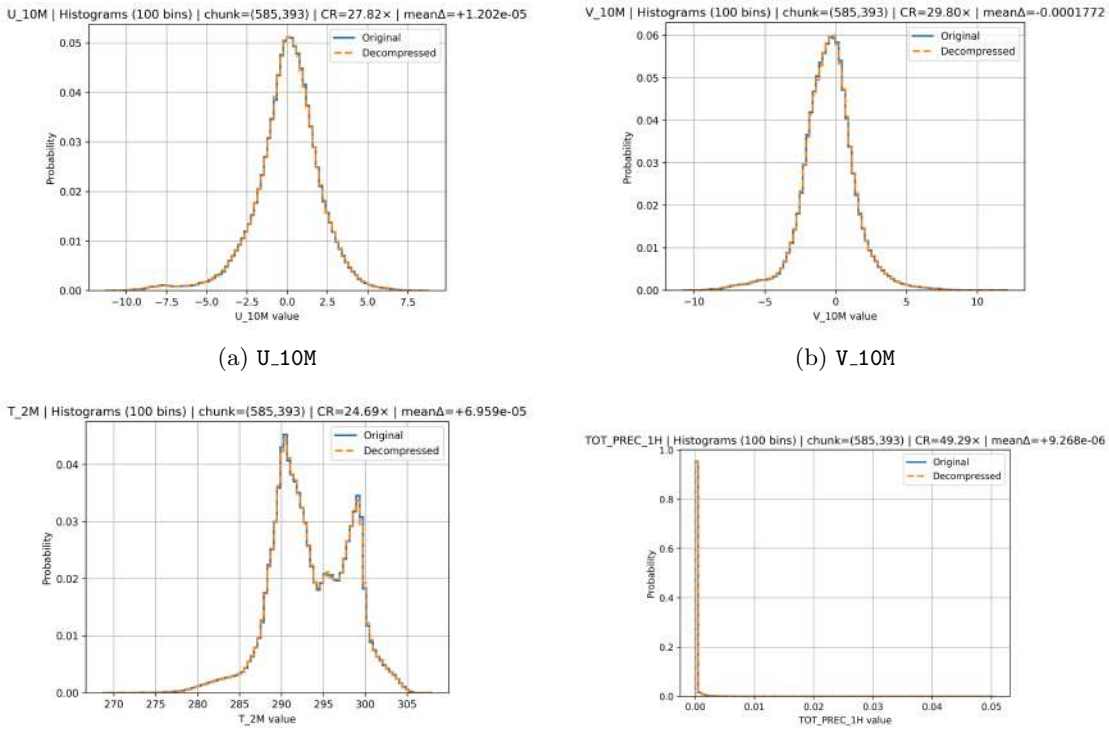
5.2 Error Distributions and Spatial Error Maps

The second level of evaluation concerns the structure of the compression error. For an original field x and decompressed field $\hat{x}$, the pointwise compression error is

$$e = \hat{x} - x. \tag{5.2}$$

In this project, the most relevant properties of e are whether it remains within the prescribed tolerance, whether it is concentrated near zero, and whether it exhibits coherent spatial artifacts.

Histograms provide a compact view of the distribution of pointwise errors. For wind and temperature, the error is evaluated directly in physical units. For precipitation, the error is evaluated in the transformed $\log(1 + x)$ space, since this is the space in which the EBCC tolerance is imposed.



(a) `U_10M` (b) `V_10M`

(c) `T_2M` (d) `TOT_PREC_1H` in log-space

Figure 5.1: Compression-error histograms for the selected variables. The precipitation error is shown in log-space because the precipitation field is transformed before compression.

The histograms in Figure 5.1 are intended to verify that the errors are concentrated around zero and remain bounded by the configured tolerances. This is an important property because a bounded but nearly uniform error distribution would still introduce substantial noise into the

reconstructed data. A distribution concentrated near zero is preferable, since most grid points then remain much closer to the original values than the worst-case tolerance.

Spatial error maps complement the histograms by showing where errors occur. This is necessary because a histogram alone does not reveal whether errors form coherent structures. In meteorological fields, spatially structured errors can be more problematic than random-looking local perturbations, since they may resemble physical features or introduce regional bias.

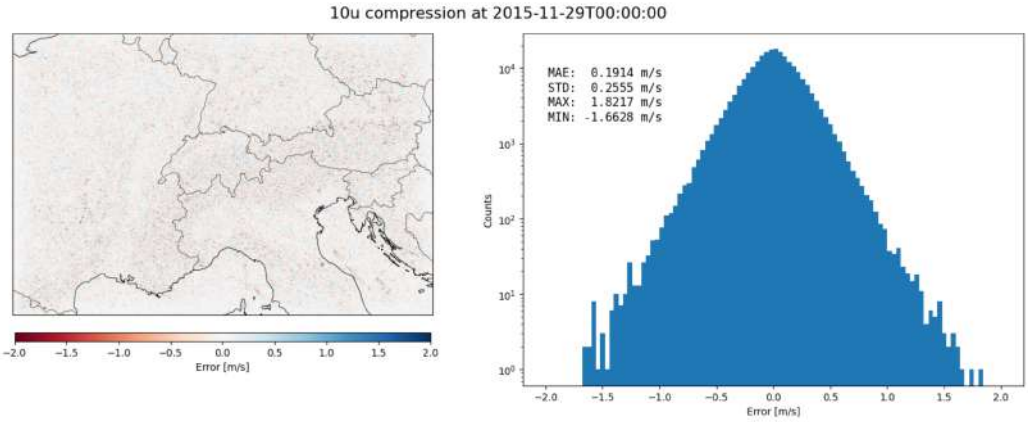


Figure 5.2: Representative spatial compression-error map for `U_10M`. The purpose of the map is to assess whether the error field remains spatially unstructured and confined within the prescribed absolute tolerance.

The representative error map in Figure 5.2 should be interpreted together with the corresponding histogram. The histogram describes the frequency of different error magnitudes, while the map verifies that errors do not organize into coherent large-scale artifacts. In the inspected examples, errors remain confined by the target tolerance and do not show evident spatial bias at the scale of the displayed fields.

5.3 Extreme-Event Case Study: Storm Petra

Beyond aggregate compression statistics, it is important to evaluate compression behavior during dynamically active periods. Extreme-event cases are useful because they contain stronger gradients, larger wind speeds, and precipitation structures that are more likely to expose weaknesses in compression. For this reason, a case study was performed for Storm Petra, starting on `2020-02-03` and covering a 72-hour period.

The case study evaluates per-frame RMSE, global RMSE, extreme reconstruction errors, and achieved compression ratios for the selected variables. The results are summarized in Tab. **??**.

Variable	Frame RMSE Min	Frame RMSE Max	Frame RMSE Avg
U_10M	0.237813	0.294940	0.267596
V_10M	0.232319	0.326553	0.280615
T_2M	0.198989	0.277003	0.249914
TOT_PREC_1H	4.33311×10^{-6}	4.37220×10^{-5}	3.10200×10^{-5}

Table 5.2: Per-frame RMSE statistics for the Storm Petra case study over a 72-hour period beginning on 2020-02-03. The precipitation quantities are reported in the transformed log-space used for compression.

Variable	Global RMSE Total	Global RMSE Avg
U_10M	19.2669	0.267838
V_10M	20.2043	0.281299
T_2M	17.9938	0.250389
TOT_PREC_1H	0.00223344	3.18927×10^{-5}

Table 5.3: Global RMSE statistics for the Storm Petra case study. The precipitation quantities are reported in the transformed log-space used for compression.

Variable	Minimum Error	Maximum Error	EBCC Error Target	Achieved CR
U_10M	-1.99833	1.99813	0.50×4	19.2752×
V_10M	-1.99859	1.99993	0.50×4	20.0371×
T_2M	-1.99924	1.99664	0.50×4	23.8280×
TOT_PREC_1H	-4.99858×10^{-4}	4.98533×10^{-4}	5.0×10^{-4}	69.75×

Table 5.4: Extreme compression errors, configured EBCC error targets, and achieved compression ratios for the Storm Petra case study. The precipitation errors and target are reported in log-space.

The wind and temperature variables remain within the expected absolute error range, with extreme errors close to but not exceeding the configured bound of approximately 2.0. The per-frame RMSE values are substantially smaller than the maximum allowed error, indicating that the prescribed tolerance is reached only by the most extreme pointwise deviations. This is consistent with the desired behavior of EBCC: the maximum error is controlled, while the typical error remains much lower.

For precipitation, the maximum positive and negative errors remain close to $\pm 5 \times 10^{-4}$ in log-space, again matching the configured target. The achieved compression ratio for precipitation during Storm Petra is lower than the aggregate precipitation compression ratio reported in Tab. 5.1, but it remains high at approximately 69.75×. This decrease is expected because an extreme-event window contains more active precipitation structures than the full dataset average.

The case-study results support the compression-level conclusion that EBCC maintains the prescribed error bounds even during a dynamically active period. This does not by itself guarantee unchanged downstream behavior, but it provides a necessary foundation for the CorrDiff evaluation in the next chapter.

5.4 Summary of Compression-Level Findings

The compression-level evaluation leads to four main findings. First, EBCC achieves substantial storage reduction on the selected COSMO-2 variables. The total size decreases from 621.640 GiB to 21.485 GiB, corresponding to an aggregate compression ratio of approximately 28.94×.

Second, compression efficiency is variable-dependent. Wind and temperature achieve compression ratios of approximately 23×, while precipitation reaches approximately 92.84× over the full selected dataset. This reflects both the sparse structure of precipitation and the use of log-space preprocessing.

Third, the representative error histograms show the expected behavior for an error-bounded compressor: errors are concentrated near zero and remain within the selected tolerance. This is important because the maximum error bound alone does not describe how typical errors are distributed.

Fourth, the Storm Petra case study confirms that the error bounds remain effective during an active meteorological event. The typical RMSE values are well below the maximum error targets, while the extreme errors remain close to the specified limits. This indicates that the compression configuration is robust at the reconstruction level for the cases inspected.

The next chapter evaluates whether these reconstruction-level findings translate into application-level reliability when the compressed data are used as input to CorrDiff.

Chapter 6

Downstream Machine-Learning Evaluation with CorrDiff

6.1 Evaluation Setup

The compression-level results in the previous chapter show that EBCC achieves large storage reduction while keeping reconstruction errors within the prescribed tolerances. This is a necessary condition for using the compressed archive, but it is not sufficient for machine-learning workflows. A downstream model may respond to small input perturbations in a nonlinear way, especially when the outputs are evaluated through extremes, thresholds, or distributional tails. For this reason, the compressed data were also evaluated through CorrDiff, a residual corrective diffusion model for kilometer-scale atmospheric downscaling [2].

The downstream comparison is organized around three fields. The first is the COSMO-2 analysis target. The second is the CorrDiff prediction obtained from original, uncompressed input data. The third is the CorrDiff prediction obtained from EBCC-compressed input data. In the figures below, the comparison is therefore consistently presented as

Target vs. Prediction from original input vs. Prediction from compressed input.

Only a representative subset of the available CorrDiff figures is included in the main text. The selected figures focus on precipitation extremes, wet-hour frequency, wind speed diagnostics, diurnal cycles, and probability-of-exceedance curves. These diagnostics were chosen because they are relevant to high-resolution weather applications and are more informative than pointwise comparisons alone. Additional precipitation, wind, diurnal-cycle, and distributional figures are collected in the appendix.

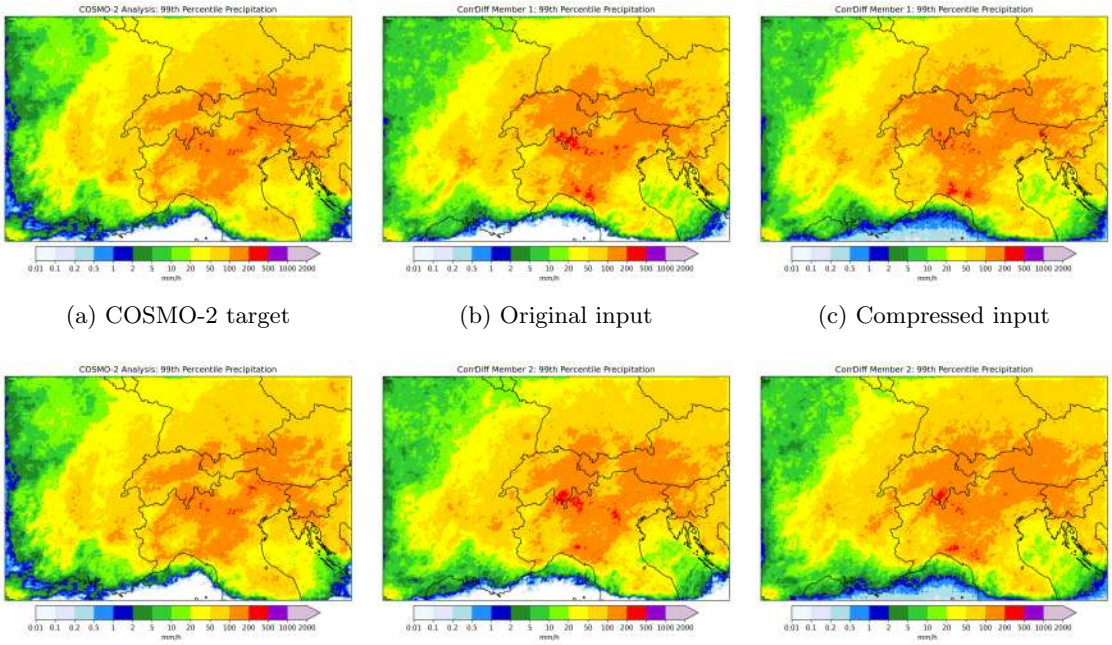
6.2 Map-Based Diagnostics

Map-based diagnostics provide a spatial view of whether compressed-input CorrDiff predictions preserve the same structures as original-input predictions. This is important because spatial agreement cannot be inferred from scalar error statistics alone. A compressed input could, in principle, preserve global distributions while shifting or distorting local structures. The maps below therefore compare the location, intensity pattern, and spatial coherence of selected precipitation and wind diagnostics.

6.2.1 Precipitation Extremes and Indices

Precipitation is one of the most sensitive variables in this project. It is sparse, highly skewed, and often evaluated through threshold-based or tail-based diagnostics. For this reason, precipitation provides a demanding test of the compressed-input workflow.

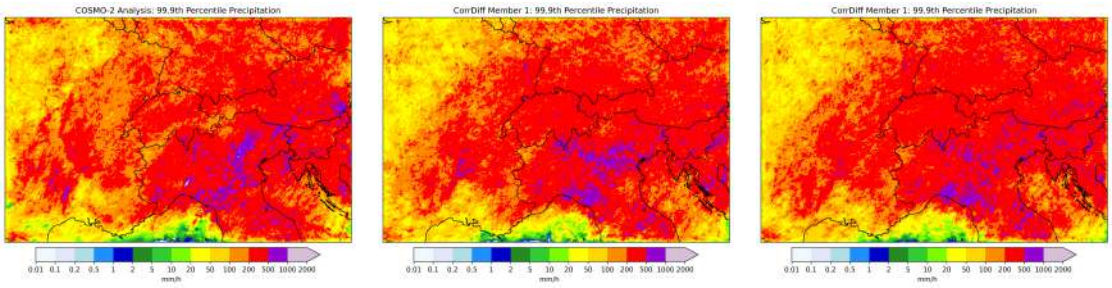
Figures 6.1 and 6.2 compare high-percentile precipitation fields from the COSMO-2 target, the CorrDiff prediction using original inputs, and the CorrDiff prediction using compressed inputs. The 99th percentile captures intense but more frequently sampled precipitation events, while the 99.9th percentile focuses on rarer extremes. In both cases, the compressed-input prediction preserves the main spatial structures visible in the original-input prediction, including the dominant regions of high precipitation intensity. Differences between the two predictions appear secondary relative to the overall spatial pattern. To complement these tail-oriented maps, additional precipitation diagnostics are included below for wet-hour frequency, mean precipitation, consecutive dry days, and consecutive wet days.



(a) COSMO-2 target (b) Original input (c) Compressed input

(d) COSMO-2 target (e) Original input (f) Compressed input

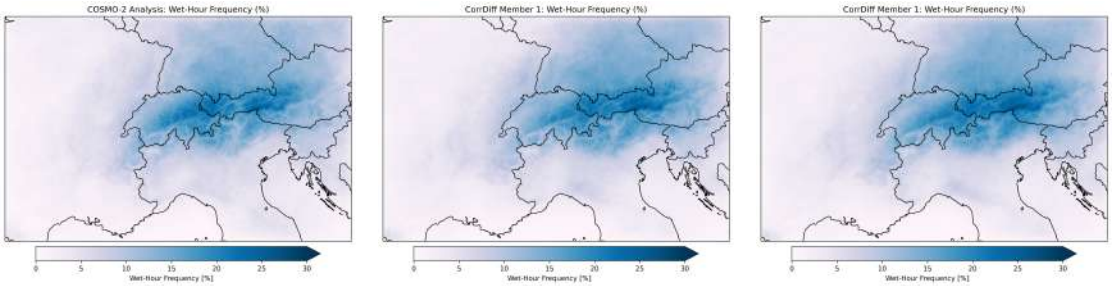
Figure 6.1: Comparison of 99th percentile precipitation for two CorrDiff members. The first column shows the COSMO-2 analysis target, while the remaining panels compare predictions from original and EBCC-compressed inputs for two different selected members. This diagnostic assesses whether high-percentile precipitation structures remain consistent across compressed-input and original-input runs.



(a) COSMO-2 target (b) CorrDiff, original input (c) CorrDiff, compressed input

Figure 6.2: Comparison of 99.9th percentile precipitation. The panels show the COSMO-2 analysis target, the CorrDiff prediction from original inputs, and the CorrDiff prediction from EBCC-compressed inputs. This figure is used as a representative precipitation-extreme diagnostic; additional percentile maps are provided in the appendix.

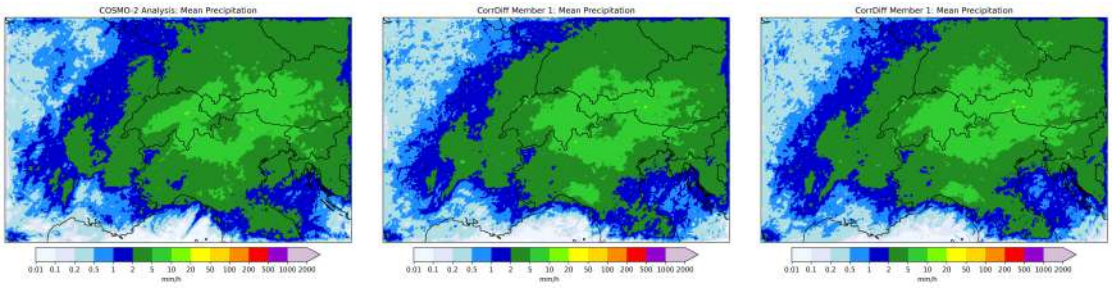
A complementary precipitation diagnostic is wet-hour frequency, shown in Figure 6.3. This quantity is sensitive to threshold crossings: small perturbations can change whether a grid point is counted as wet when precipitation is close to the threshold. The close agreement between the original-input and compressed-input predictions indicates that the EBCC perturbations do not substantially alter this threshold-based diagnostic in the selected evaluation.



(a) COSMO-2 target (b) CorrDiff, original input (c) CorrDiff, compressed input

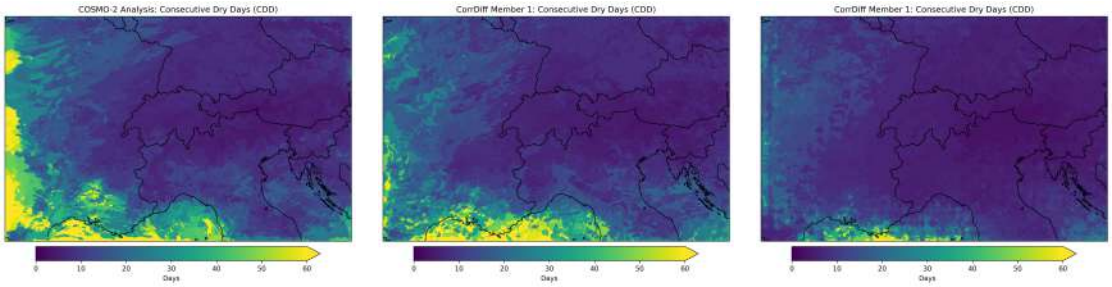
Figure 6.3: Wet-hour frequency comparison. This diagnostic is threshold-based and therefore tests whether compression-induced perturbations alter the occurrence frequency of wet conditions. The compressed-input prediction remains visually close to the original-input prediction.

Mean precipitation, consecutive dry days, and consecutive wet days provide complementary checks on accumulated and persistence-based precipitation behavior. While percentile maps focus on the upper tail and wet-hour frequency focuses on threshold occurrence, these diagnostics test whether the compressed-input workflow preserves broader precipitation structure and temporal persistence.



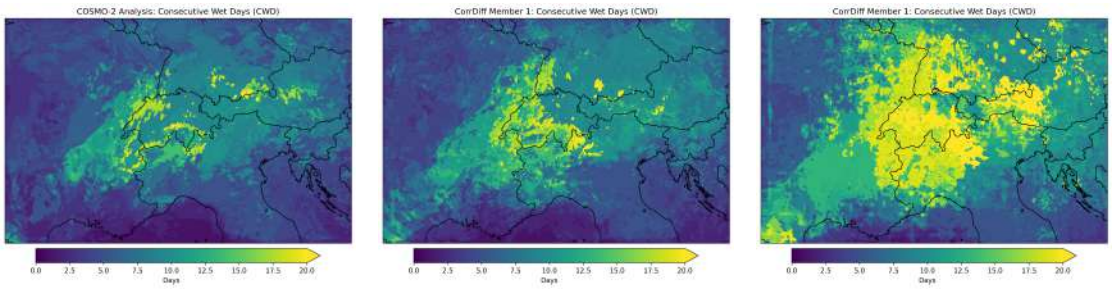
(a) COSMO-2 target (b) CorrDiff, original input (c) CorrDiff, compressed input

Figure 6.4: Comparison of mean precipitation. This diagnostic evaluates whether EBCC-compressed inputs preserve the broader spatial structure of accumulated precipitation in the CorrDiff output, beyond the high-percentile and threshold-based diagnostics.



(a) COSMO-2 target (b) CorrDiff, original input (c) CorrDiff, compressed input

Figure 6.5: Comparison of consecutive dry days. This persistence-based diagnostic tests whether the compressed-input CorrDiff run preserves the spatial pattern of dry-period duration relative to the original-input run.



(a) COSMO-2 target (b) CorrDiff, original input (c) CorrDiff, compressed input

Figure 6.6: Comparison of consecutive wet days. This persistence-based diagnostic complements wet-hour frequency by assessing whether the temporal clustering of wet conditions is preserved when CorrDiff is driven by EBCC-compressed inputs.

The mean precipitation and wet/dry persistence maps strengthen the conclusion from the percentile and wet-hour diagnostics: the compressed-input CorrDiff run preserves not only pre-

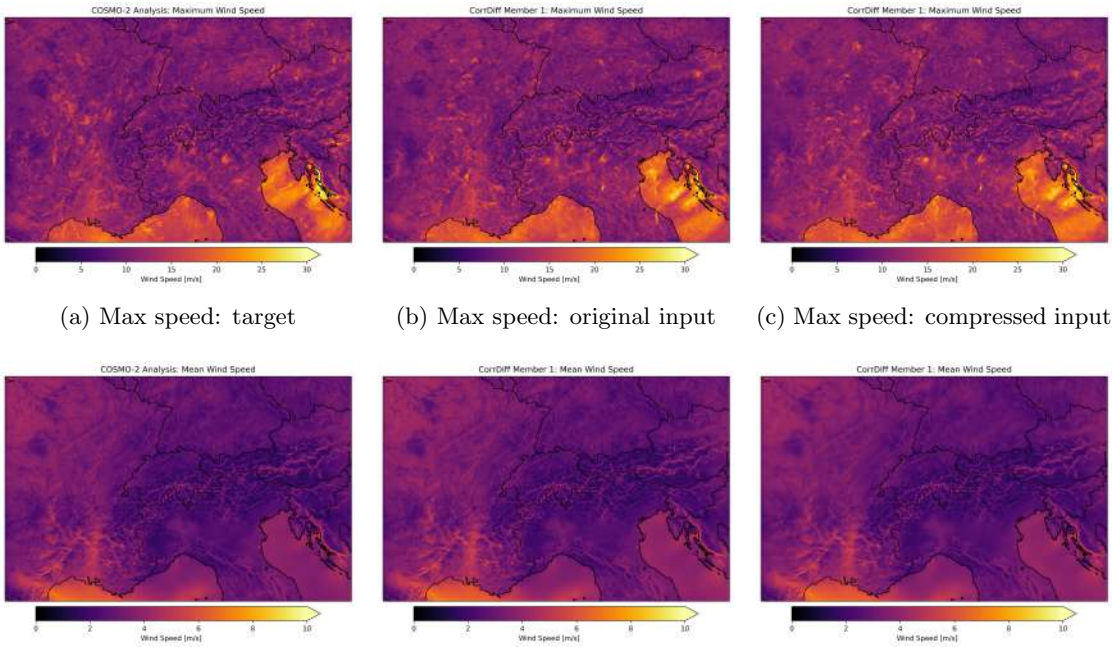
cipitation intensity patterns, but also broader accumulation and event-persistence structures in the selected evaluation.

Together, the precipitation maps suggest that the compressed-input CorrDiff run preserves tail-oriented, threshold-oriented, accumulated, and persistence-based precipitation structures in the selected examples. Further precipitation diagnostics, including the 99.99th percentile map and maximum accumulation indices, are included in the appendix.

6.2.2 Wind Diagnostics

Wind diagnostics provide a second major component of the downstream evaluation. Since EBCC was applied independently to the two horizontal wind components, it is important to verify that derived wind quantities remain consistent after CorrDiff inference. Two representative examples are shown here: maximum wind speed and mean wind speed.

Figure 6.7 displays the target, original-input prediction, and compressed-input prediction for both maximum and mean wind speed. The maximum wind speed diagnostic is sensitive to local extremes, while the mean wind speed diagnostic emphasizes broader spatial structure. In both cases, the compressed-input prediction remains close to the original-input prediction in the displayed patterns.



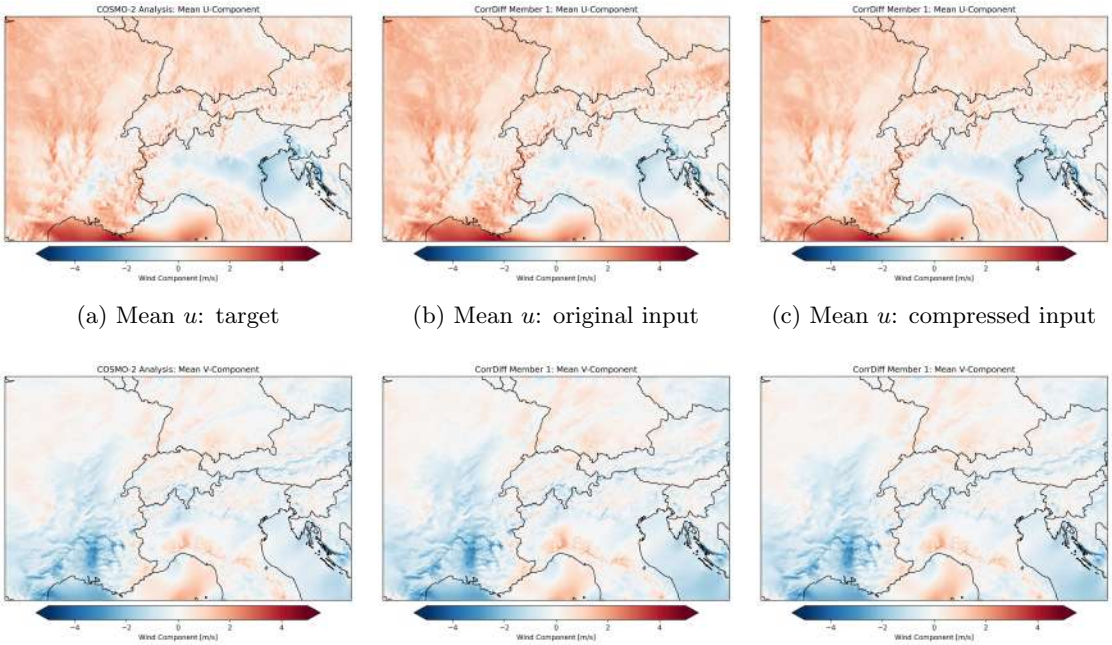
(a) Max speed: target (b) Max speed: original input (c) Max speed: compressed input

(d) Mean speed: target (e) Mean speed: original input (f) Mean speed: compressed input

Figure 6.7: Wind-speed diagnostics from CorrDiff. The first row compares maximum wind speed, while the second row compares mean wind speed. In both diagnostics, the compressed-input prediction preserves the main spatial structures of the original-input prediction.

Mean wind speed summarizes the magnitude of the horizontal wind field, but it does not show whether the directional components are preserved separately. Since EBCC is applied independently to `10u` and `10v`, the mean zonal and meridional wind components provide a more

direct component-wise diagnostic. Figure 6.8 compares these two diagnostics for the COSMO-2 target, the original-input CorrDiff prediction, and the compressed-input CorrDiff prediction.



(a) Mean u: target (b) Mean u: original input (c) Mean u: compressed input

(d) Mean v: target (e) Mean v: original input (f) Mean v: compressed input

Figure 6.8: Mean horizontal wind-component diagnostics from CorrDiff. The first row compares the mean zonal component u, while the second row compares the mean meridional component v. These component-wise diagnostics directly test whether the independently compressed wind inputs preserve the directional structure of the CorrDiff output.

These maps indicate that the compression of the wind components does not introduce a visible degradation in the selected wind-speed or component-wise diagnostics after CorrDiff inference. Additional wind diagnostics, including calm frequency, breeze categories, wind direction variability, prevailing direction, and wind power density, are reported in the appendix.

6.3 Diurnal-Cycle Diagnostics

Map-based diagnostics show spatial structure, but they do not directly address temporal behavior. Diurnal cycles provide a compact way to assess whether the daily phase and amplitude of predicted variables are preserved. This is particularly important for near-surface quantities such as 2 m temperature, wind speed, precipitation amount, and wet-hour frequency.

The available diurnal-cycle figures compare the behavior of CorrDiff predictions generated from original and compressed inputs. Figure 6.9 and Figure 6.10 show the 2 m temperature and wind-speed cycles. These diagnostics test whether compression affects systematic daily variability, including peak timing and amplitude.

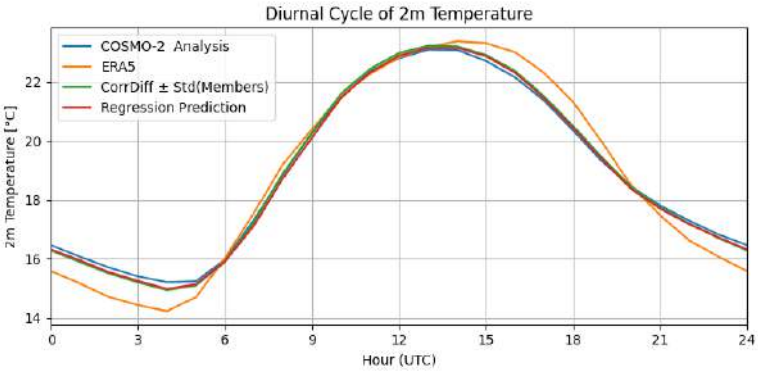


Figure 6.9: Diurnal cycle of 2 m temperature from the CorrDiff evaluation. The figure compares the behavior obtained from original and EBCC-compressed inputs and assesses whether the daily phase and amplitude of near-surface temperature are preserved.

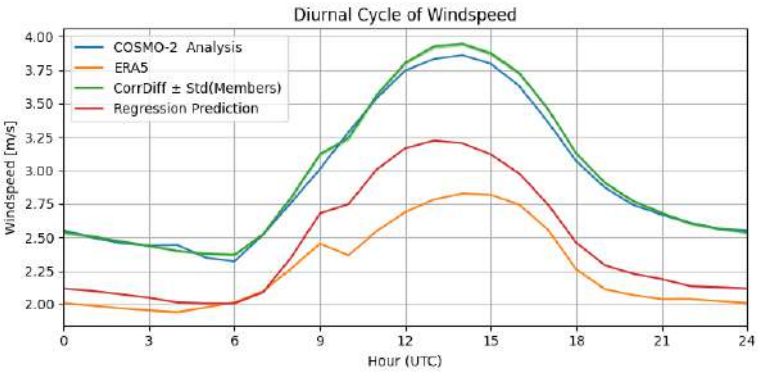


Figure 6.10: Diurnal cycle of wind speed from the CorrDiff evaluation. The comparison tests whether EBCC-compressed inputs preserve the daily variability of predicted near-surface wind speed.

Precipitation-related diurnal cycles are more sensitive than temperature and wind because precipitation is intermittent and strongly influenced by threshold and tail behavior. The selected diagnostics include precipitation amount, wet-hour frequency, and the 99th percentile precipitation cycle.

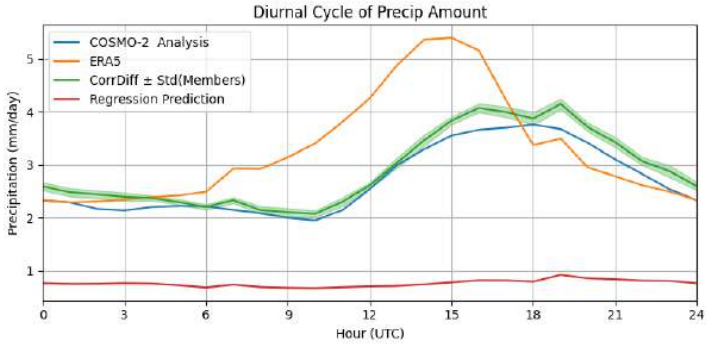


Figure 6.11: Diurnal cycle of precipitation amount from the CorrDiff evaluation. This diagnostic assesses whether EBCC-compressed inputs preserve the timing and magnitude of the predicted daily precipitation cycle.

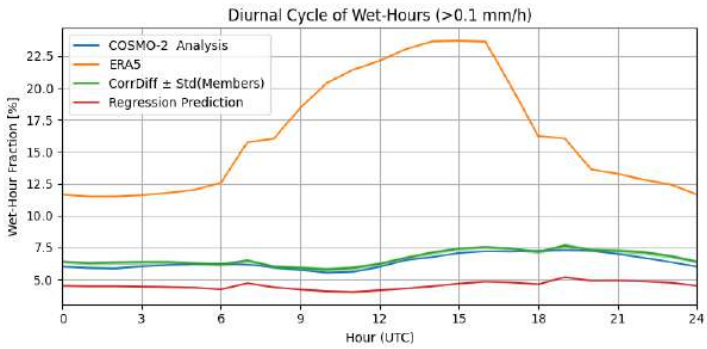


Figure 6.12: Diurnal cycle of wet-hour frequency from the CorrDiff evaluation. Since wet-hour frequency is threshold-based, this figure provides a sensitive test of whether compression-induced input perturbations alter the occurrence timing of wet conditions.

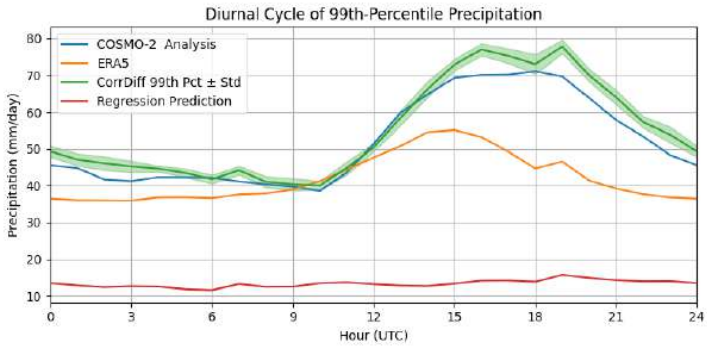


Figure 6.13: Diurnal cycle of 99th percentile precipitation from the CorrDiff evaluation. This tail-oriented diagnostic tests whether high-percentile precipitation behavior is preserved when CorrDiff is driven by EBCC-compressed inputs.

The diurnal-cycle diagnostics support the same interpretation as the map-based diagnostics: the compressed-input workflow does not visibly alter the principal temporal structures in the selected results. The remaining diurnal-cycle figures are included in the appendix for completeness.

6.4 Distributional and Tail Diagnostics

Distributional diagnostics aggregate information over many grid points and times. They are useful because two fields may look similar in a few selected maps while still differing in their pooled distributions. In this project, the main distributional checks focus on precipitation over land and probability-of-exceedance curves.

Let X denote a pooled sample of a diagnostic quantity, such as hourly precipitation over land or wind speed. The probability of exceedance at threshold a is

$$P(X > a). \tag{6.1}$$

This diagnostic is especially relevant for tail behavior because it directly measures how frequently high values occur.

Figure 6.14 and Figure 6.15 show the pooled precipitation distribution and precipitation probability-of-exceedance diagnostic. These plots test whether the compressed-input CorrDiff run preserves the distribution of precipitation values, especially away from the center of the distribution.

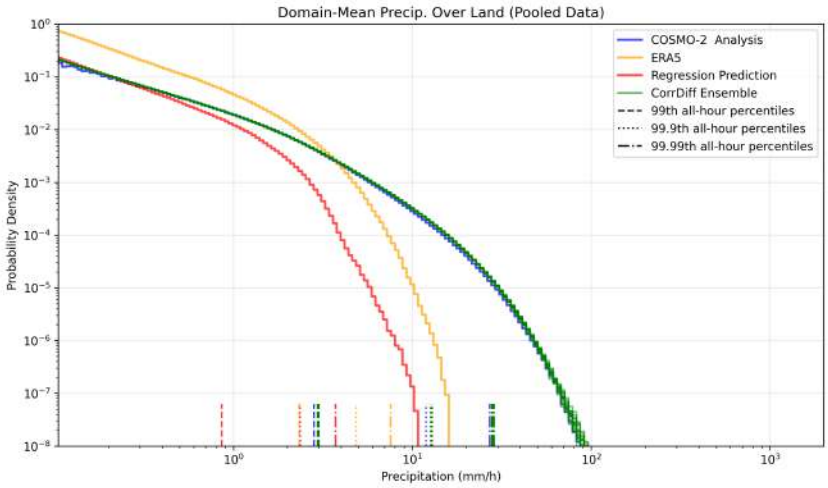


Figure 6.14: Pooled precipitation distribution over land. This diagnostic assesses whether the compressed-input CorrDiff prediction preserves the aggregate precipitation distribution beyond individual map-based examples.

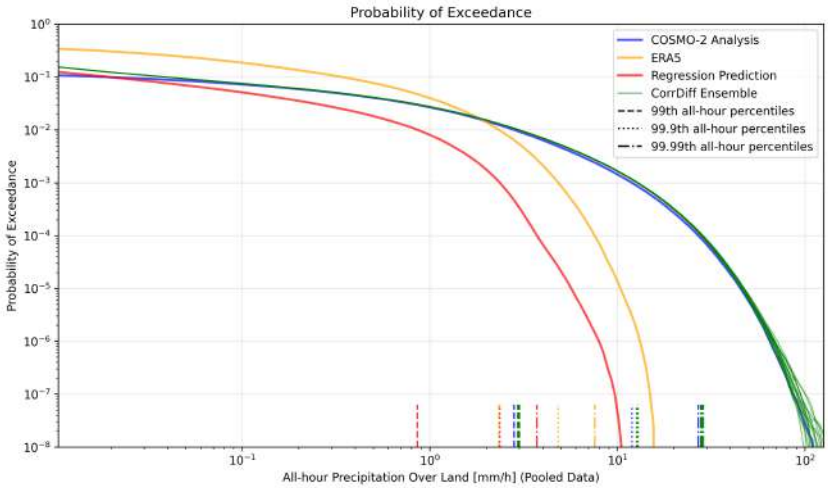


Figure 6.15: Probability of exceedance for precipitation over land. The curve provides a direct tail diagnostic by comparing how frequently precipitation values exceed a range of thresholds.

Wind-related exceedance diagnostics are shown in Figures 6.16 to 6.18. These curves evaluate not only wind speed, but also the absolute values of the horizontal wind components. This is useful because the compressed inputs are applied to `10u` and `10v` separately, while many downstream quantities depend on their combined or component-wise behavior.

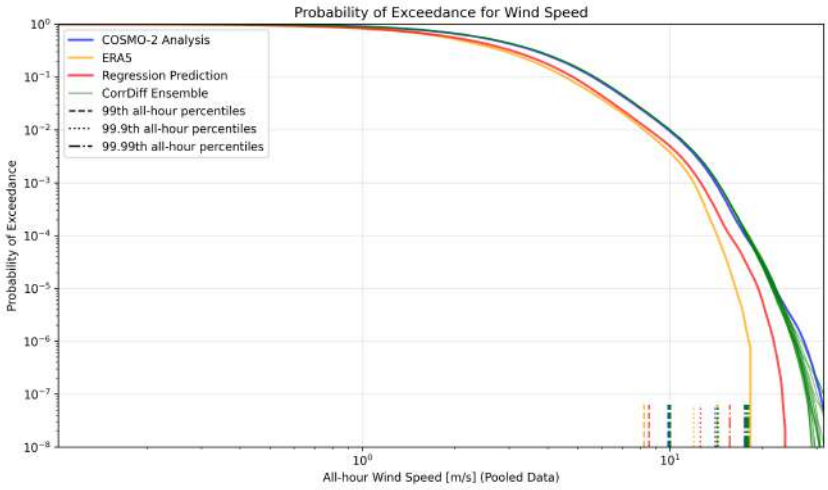


Figure 6.16: Probability of exceedance for wind speed. This diagnostic assesses whether the compressed-input CorrDiff prediction preserves the tail behavior of derived wind speed.

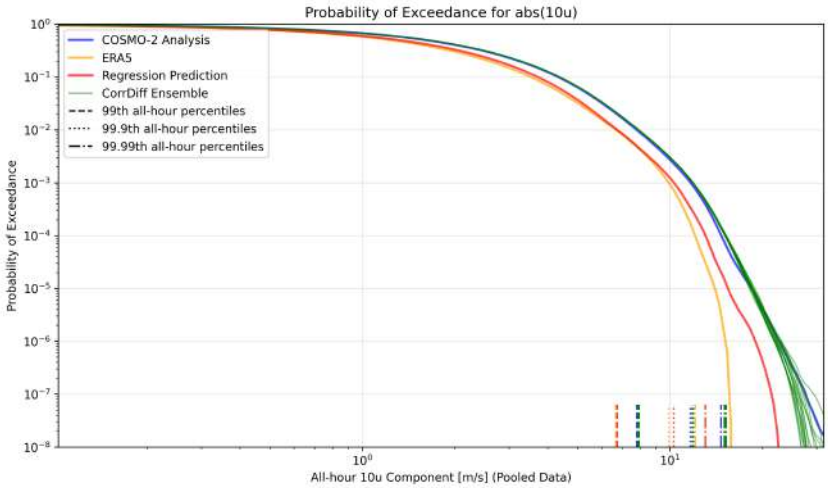


Figure 6.17: Probability of exceedance for $|10u|$. This component-wise diagnostic tests whether the zonal wind contribution remains distributionally consistent after using EBCC-compressed inputs.

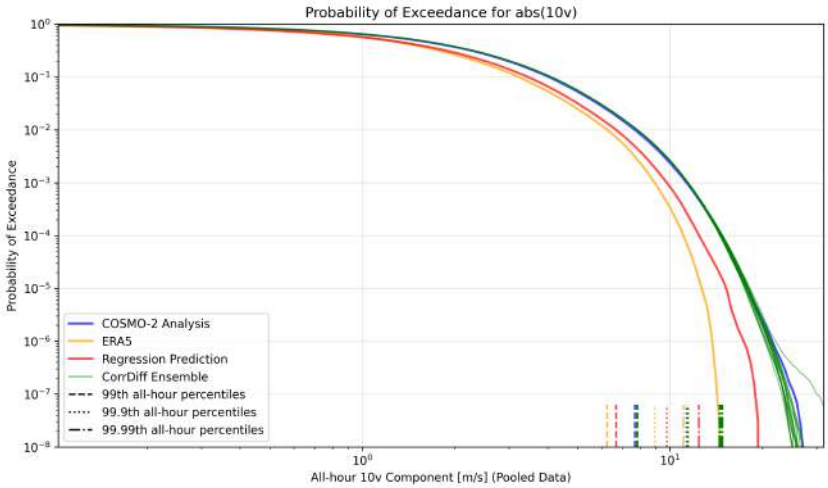


Figure 6.18: Probability of exceedance for $|10v|$. This component-wise diagnostic tests whether the meridional wind contribution remains distributionally consistent after using EBCC-compressed inputs.

The distributional diagnostics are important because they test the compressed-input workflow at a different level from spatial maps. The selected plots indicate that the compressed-input predictions remain close to the original-input behavior across pooled precipitation and wind-tail diagnostics.

6.5 Summary of CorrDiff Findings

The CorrDiff evaluation provides an application-level assessment of the EBCC-compressed archive. Across the selected diagnostics, the compressed-input predictions remain visually and distributionally close to the original-input predictions. This holds for representative precipitation extremes, wet-hour frequency, maximum and mean wind speed, diurnal cycles, and probability-of-exceedance curves.

The precipitation diagnostics are particularly important because precipitation is sparse, skewed, and sensitive to threshold and tail behavior. The selected maps and distributional plots indicate that the use of compressed inputs does not materially change the main precipitation structures in the tested setting. Similarly, the wind diagnostics suggest that compression of the two horizontal wind components does not introduce visible degradation in derived wind-speed quantities after CorrDiff inference.

These findings should be interpreted within the scope of the experiment. The evaluation uses selected variables, selected tolerances, selected diagnostics, and the available CorrDiff outputs. It does not prove that all downstream machine-learning models or all rare meteorological events are insensitive to compression. However, it does provide evidence that, under the tested configuration, EBCC-compressed COSMO-2 inputs preserve the behavior of a representative high-resolution machine-learning downscaling workflow.

Chapter 7

Discussion, Conclusion, and Outlook

7.1 Interpretation of Compression and CorrDiff Results

The results of this project support the use of EBCC as a practical error-bounded compression method for the selected COSMO-2 analysis variables. At the compression level, the archive size was reduced from 621.640 GiB to 21.485 GiB, corresponding to an aggregate compression ratio of approximately 28.94×. The achieved reduction is substantial, especially because it is obtained under explicit error constraints rather than through an uncontrolled lossy compression setting.

The per-variable compression ratios show that compression performance depends strongly on the structure of the field. Wind components and temperature achieve similar compression ratios, while hourly precipitation reaches a much larger ratio. This is consistent with the sparsity and near-zero dominance of precipitation fields, as well as the use of the $\log(1 + x)$ preprocessing transform. The precipitation result is therefore not directly comparable to the wind and temperature results in physical space, but it demonstrates that variable-specific preprocessing can be important for effective compression.

The error statistics and histograms indicate that EBCC behaves as intended in this setting. Reconstruction errors remain bounded by the prescribed tolerances and are concentrated near zero rather than being spread uniformly across the admissible error interval. This is relevant because the maximum error bound alone only controls the worst local error; it does not guarantee that typical errors are small. The inspected spatial error maps further suggest that the representative compression errors do not form coherent large-scale artifacts.

The Storm Petra case study provides an additional check under dynamically active conditions. The extreme errors remain close to the configured bounds, while the RMSE values are substantially smaller than the maximum allowed deviations. This indicates that the selected tolerances are reached only by the largest pointwise errors, whereas the typical reconstruction error remains lower. The case study therefore supports the conclusion that the compression configuration is robust at the reconstruction level for the inspected event window.

The CorrDiff evaluation extends the analysis from reconstruction quality to downstream usability. Across the selected map-based, diurnal-cycle, and distributional diagnostics, the predictions produced from EBCC-compressed inputs remain close to the predictions produced from original inputs. This is particularly important for precipitation extremes, wet-hour frequency, wind-speed diagnostics, and probability-of-exceedance curves, because these quantities are more

sensitive than simple mean fields. Within the tested setting, the compressed inputs preserve the main spatial patterns, temporal cycles, and distributional tail behavior of the CorrDiff outputs.

Taken together, the results suggest that EBCC compression can be compatible with downstream machine-learning workflows when tolerances are chosen carefully and evaluated at both the input and application levels. The main conclusion is not that compression is universally harmless, but that the tested EBCC configuration preserves the behavior of the selected CorrDiff diagnostics to a degree that is sufficient for this project.

7.2 Operational Advantages

The first operational advantage of EBCC is explicit error control. The user specifies tolerances for the compressed variables, and the method aims to satisfy these bounds through its residual correction mechanism. This is preferable to using a generic lossy compressor without a clear reconstruction guarantee, especially when the compressed data will be used for scientific or operational diagnostics.

A second advantage is the treatment of long-tail errors. Weather and climate applications can be sensitive to rare large deviations, particularly in extremes and threshold-based diagnostics. EBCC addresses this issue through its residual layer, which corrects excessive errors left by the base compression layer. This makes the method well suited to applications where both high compression and controlled local errors are required.

A third advantage is compatibility with array-based scientific workflows. In this project, EBCC was integrated into a Zarr-based pipeline by compressing each variable independently and using one full horizontal field per chunk. This preserves a natural data organization and keeps the compressed archive aligned with common analysis workflows based on time-indexed gridded fields.

The fourth advantage is parallelism. Since each time index and variable frame can be compressed independently, the workflow is embarrassingly parallel over time. This made it possible to use a simple SLURM-based strategy in which many tasks compress different time indices concurrently. Such a structure is appropriate for high-performance computing environments, where parallel throughput is often more important than optimizing a single serial compression process.

Finally, the storage reduction itself is operationally important. Reducing the selected archive by more than an order of magnitude lowers storage pressure and can reduce data movement in downstream workflows. For machine-learning applications, where datasets may be read repeatedly during evaluation or training, this reduction can be valuable even when compression is performed offline.

7.3 Limitations

Several limitations should be kept in mind when interpreting the results. First, the selected error tolerances were chosen empirically. They worked well for the variables, time window, and diagnostics considered here, but they should not be interpreted as universally optimal. Different applications may require stricter or looser tolerances depending on the physical variable and the downstream use case.

Second, precipitation was compressed in log-space. This is appropriate for handling its skewed and sparse distribution, but it means that the prescribed error bound is not a direct absolute error bound in precipitation units. As a result, physical-space precipitation errors require separate interpretation after applying the inverse transformation.

Third, the compression was applied independently to each variable. This is convenient and allows variable-specific tolerances, but it does not explicitly enforce cross-variable consistency during compression. In the present project, such consistency was assessed indirectly through CorrDiff diagnostics. Future work could examine whether multivariate constraints or joint diagnostics reveal additional sensitivities.

Fourth, the downstream evaluation was limited to the available CorrDiff outputs and selected diagnostics. The results show strong agreement for the inspected cases, but they do not prove that all downstream models, all meteorological regimes, or all rare events would be equally insensitive to compression. A broader evaluation would require additional periods, additional events, more ensemble members, and possibly other ML models.

Fifth, the implementation was designed as a project pipeline rather than a fully optimized production tool. The parallelization strategy is simple and effective, but the realized performance may depend on filesystem behavior, concurrent write patterns, metadata overhead, and chunk layout. These aspects would need further testing before operational deployment at larger scale.

7.4 Recommendations for Future Use

Future use of EBCC in similar workflows should begin with variable-specific tolerance selection. The tolerance should not be chosen only from the desired compression ratio, but from the sensitivity of the downstream diagnostics. For example, precipitation extremes, threshold frequencies, and probability-of-exceedance curves should be tested explicitly when precipitation is an important variable.

It is also advisable to evaluate compression in two stages. The first stage should inspect reconstruction-level quantities such as maximum error, RMSE, error histograms, and spatial error maps. The second stage should evaluate downstream diagnostics after the compressed data have been used by the target workflow. This distinction is important because small input perturbations may have different effects depending on the model and diagnostic.

For precipitation, the log-space transform should be documented carefully whenever compressed data are shared or reused. Users should be aware that the error target is imposed in transformed space, and any physical-space interpretation should account for the inverse transform.

For larger applications, the chunking strategy should be selected jointly with access patterns. The frame-wise chunking used here is appropriate for time-parallel compression and full-field access, but other workflows may require smaller spatial chunks or different layouts. The best configuration depends on whether the archive is mainly used for sequential time processing, random spatial access, or machine-learning batch loading.

Finally, future evaluations should include more diverse meteorological situations. Extreme events, seasonal variation, different precipitation regimes, and longer time windows would provide a stronger basis for operational tolerance recommendations.

7.5 Concluding Remarks

This semester project implemented and evaluated an EBCC-based compression workflow for selected MeteoSwiss COSMO-2 analysis variables. The project demonstrated that EBCC can be integrated into a Zarr-based data pipeline, configured with variable-specific error tolerances, and parallelized over time indices on an HPC system.

The compression results show a substantial storage reduction, with an aggregate compression ratio of approximately 28.94× for the selected variables and time window. The reconstruction

errors remain bounded by the prescribed tolerances and are concentrated near zero in the representative diagnostics. The Storm Petra case study further indicates that the error bounds remain effective during an active meteorological period.

The downstream CorrDiff evaluation shows that compressed-input predictions remain close to original-input predictions across the selected map-based, diurnal-cycle, and distributional diagnostics. This provides evidence that EBCC-compressed inputs can preserve the behavior of a representative machine-learning downscaling workflow under the tested conditions.

Overall, the project supports the view that error-bounded compression can be a practical component of high-resolution weather-data workflows, provided that tolerance selection is performed carefully and evaluated with respect to the intended downstream application.

7.6 Future Work

Several extensions are natural. First, the tolerance-selection procedure should be made more systematic. Instead of selecting tolerances empirically, future work could sweep across several error targets and quantify the resulting trade-off between compression ratio, reconstruction error, and downstream diagnostic sensitivity.

Second, the CorrDiff evaluation could be expanded to include more ensemble members and longer evaluation periods. This would allow a more robust assessment of whether compressed inputs preserve not only individual deterministic predictions, but also ensemble statistics and uncertainty-related quantities.

Third, additional meteorological variables could be compressed and evaluated. The present project focused on near-surface wind, temperature, and precipitation, but operational and research workflows often require pressure-level variables, humidity, radiation, and other diagnostic fields. Each variable may require a different tolerance and possibly different preprocessing.

Fourth, future work could compare EBCC directly with alternative compressors under the same COSMO-2 and CorrDiff evaluation setup. Such a comparison would help determine whether EBCC's long-tail error control leads to measurable downstream advantages in this specific application.

Fifth, performance and I/O behavior should be studied more systematically. This includes compression throughput, decompression throughput, metadata overhead, scaling with task count, and interaction with the storage system. Such measurements would be needed before integrating the workflow into a larger operational setting.

7.7 Acknowledgments

I would like to express my sincere gratitude to Dr. Xavier Lapillonne for organizing the project, welcoming me at MeteoSwiss, and providing continuous supervision throughout the semester. I am equally grateful to Dr. Francesco Zanetta for his extensive technical support, guidance in implementation, help with the workflow, and feedback on the final results.

I would also like to thank Prof. Dr. Torsten Hoefler, Langwen Huang, Jiayong Li, and the SPCL team at ETH Zurich for developing EBCC and for their technical support during the project. Their work provided the methodological and software foundation for this study.

I am grateful to Petar Stamenkovic from MeteoSwiss for assistance in the generating the CorrDiff evaluation outputs and figures used in the downstream analysis. Finally, I thank MeteoSwiss for hosting the project and enabling regular in-person collaboration, and ETH Zurich for the academic framework within which this semester project was carried out.

Appendix A

Additional CorrDiff Figures

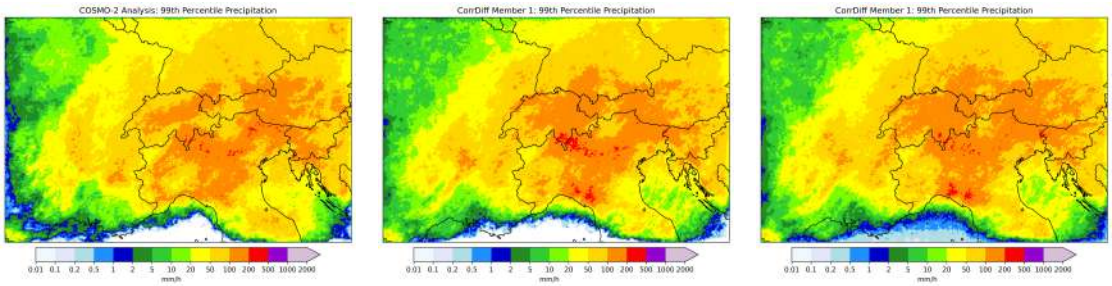
This appendix collects the CorrDiff diagnostic figures used throughout the downstream evaluation, including figures already shown in the main text. The purpose is to provide a compact visual catalog of the map-based, diurnal-cycle, and distributional diagnostics. Unless otherwise stated, map-based comparisons are shown as COSMO-2 analysis target, CorrDiff prediction from original inputs, and CorrDiff prediction from EBCC-compressed inputs.

For compactness and consistency, the map-based comparisons generally use only CorrDiff prediction member 00. Additional available outputs, including other prediction members, regression predictions, and baseline fields (ERA5), are not displayed systematically here; they are retained in the figure directory.

A.1 Map-Based Diagnostics

A.1.1 Precipitation Extremes

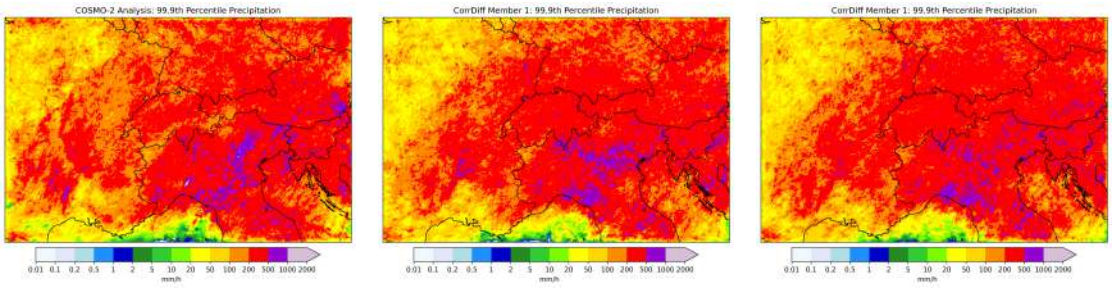
99th Percentile Precipitation



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.1: Comparison of 99th percentile precipitation. This diagnostic captures intense events and tests whether high-percentile structures remain consistent when CorrDiff uses EBCC-compressed inputs.

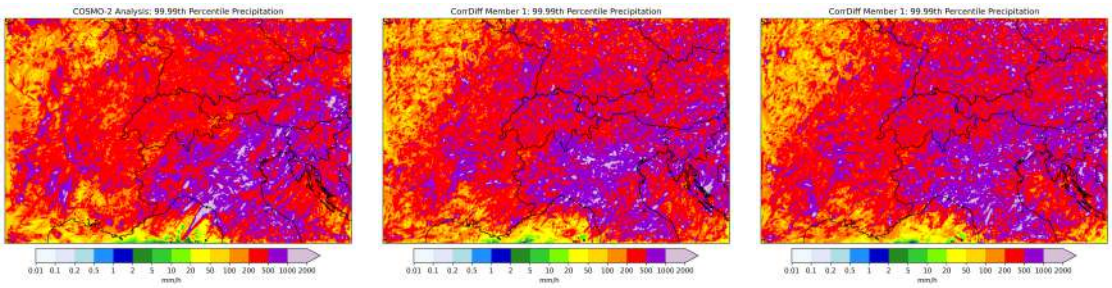
99.9th Percentile Precipitation



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.2: Comparison of 99.9th percentile precipitation. This diagnostic focuses on rarer precipitation extremes than the 99th percentile map and provides a tail-oriented test of the compressed-input CorrDiff workflow.

99.99th Percentile Precipitation

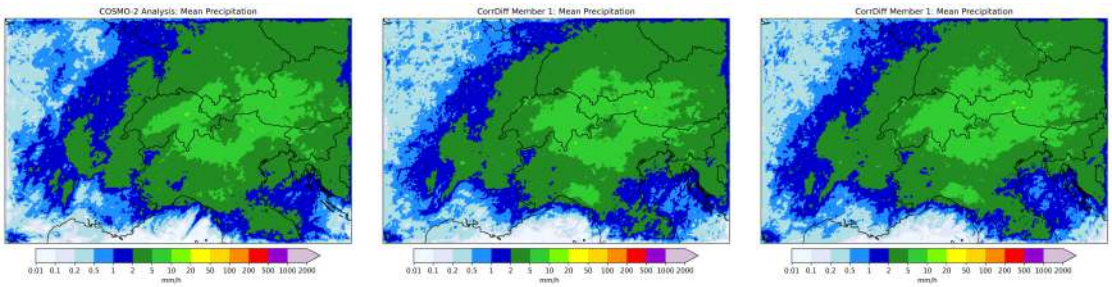


(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.3: Comparison of 99.99th percentile precipitation. This is the most tail-focused precipitation percentile diagnostic included in the figure set and is useful for inspecting the rarest precipitation intensities.

A.1.2 Accumulated and Threshold-Based Precipitation Indices

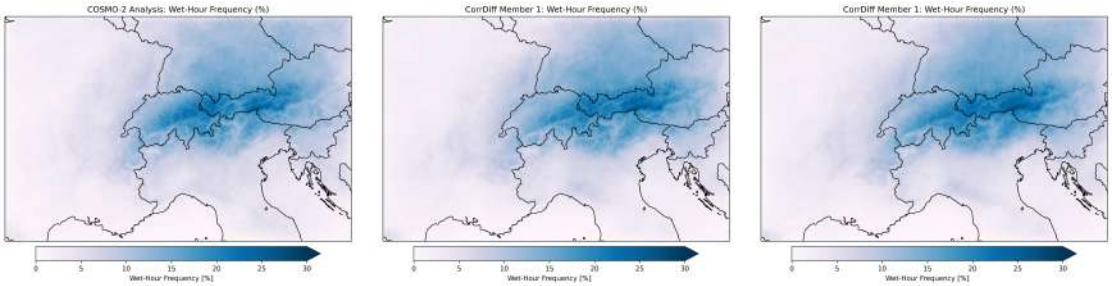
Mean Precipitation



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.4: Comparison of mean precipitation. This diagnostic evaluates whether the compressed-input CorrDiff run preserves the broader accumulated precipitation structure, beyond tail-focused percentile diagnostics.

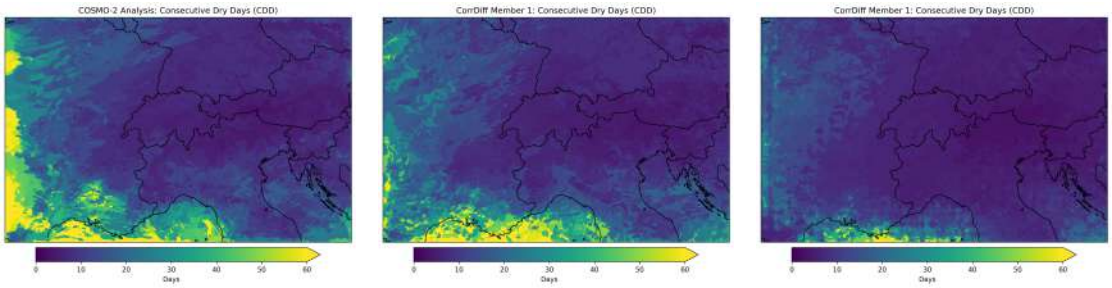
Wet-Hour Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.5: Comparison of wet-hour frequency. This threshold-based diagnostic tests whether the occurrence frequency of wet conditions is preserved under compressed-input CorrDiff inference.

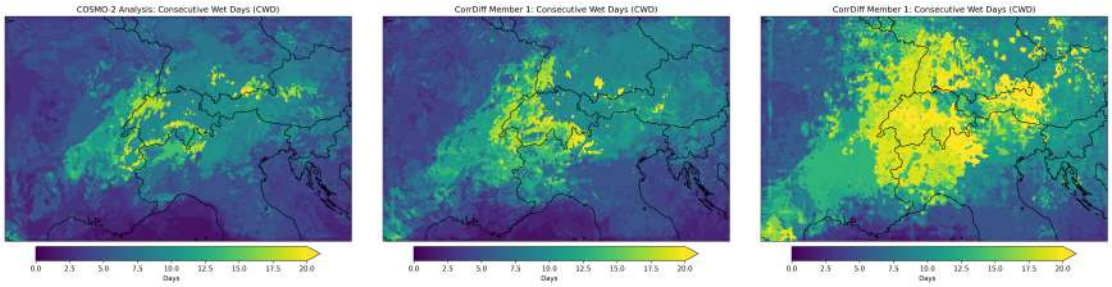
Consecutive Dry Days



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.6: Comparison of consecutive dry days. This persistence-based precipitation diagnostic evaluates whether dry-period duration patterns are preserved in the compressed-input CorrDiff prediction.

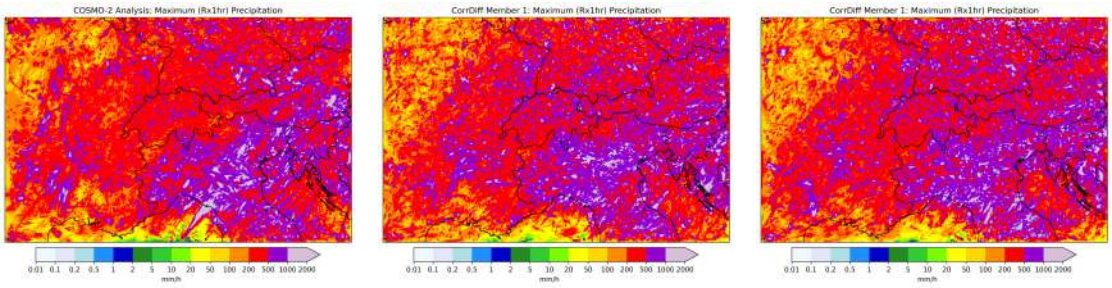
Consecutive Wet Days



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.7: Comparison of consecutive wet days. This persistence-based diagnostic complements wet-hour frequency by assessing the temporal clustering of wet conditions.

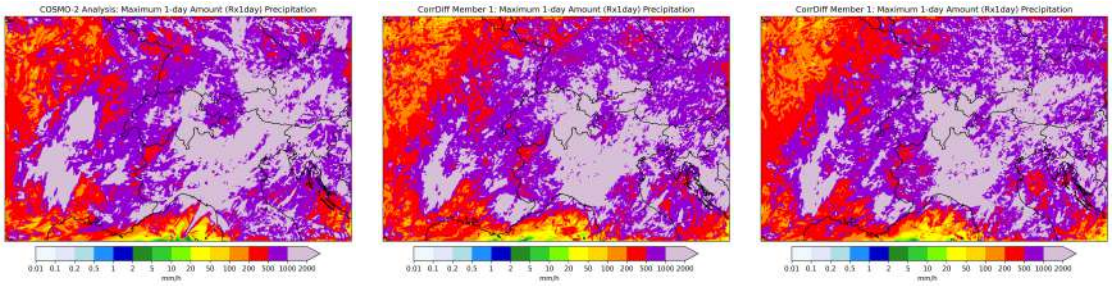
Maximum 1-Hour Precipitation



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.8: Comparison of maximum 1-hour precipitation, Rx1h. This diagnostic evaluates whether short-duration precipitation extremes are preserved when CorrDiff is driven by EBCC-compressed inputs.

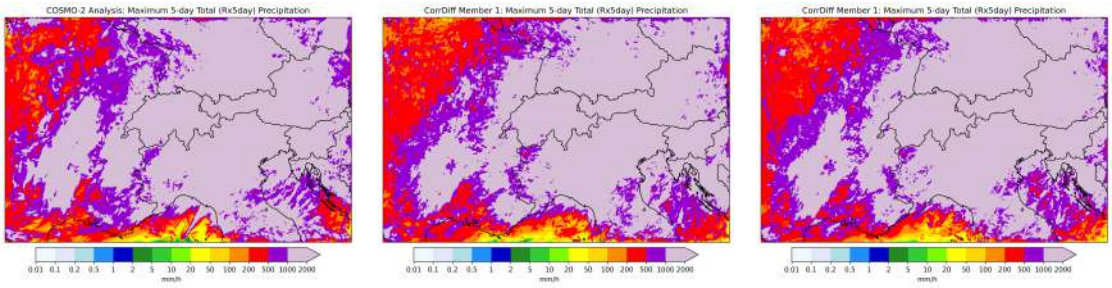
Maximum 1-Day Precipitation



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.9: Comparison of maximum 1-day precipitation, Rx1day. This diagnostic assesses whether daily accumulated precipitation extremes remain consistent between original-input and compressed-input CorrDiff predictions.

Maximum 5-Day Precipitation

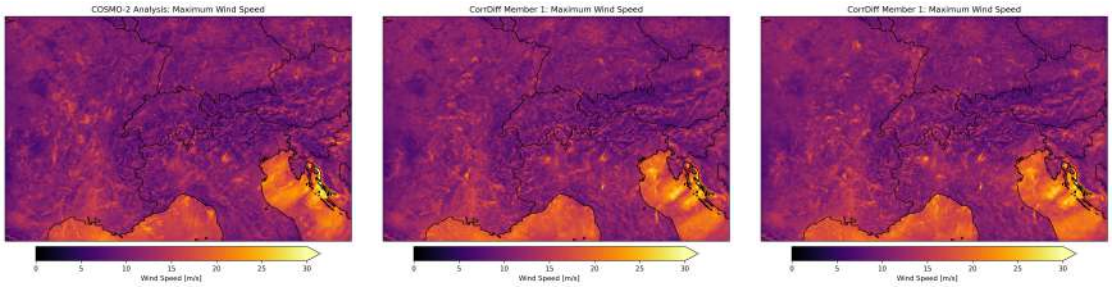


(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.10: Comparison of maximum 5-day precipitation, Rx5day. This diagnostic evaluates whether multi-day accumulated precipitation extremes are preserved under compressed-input inference.

A.1.3 Wind Speed Diagnostics

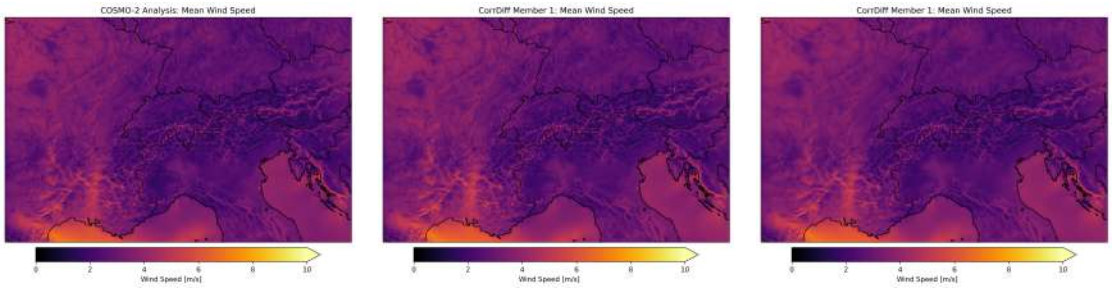
Maximum Wind Speed



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.11: Comparison of maximum wind speed. This diagnostic evaluates whether local wind-speed extremes are preserved after using EBCC-compressed inputs.

Mean Wind Speed

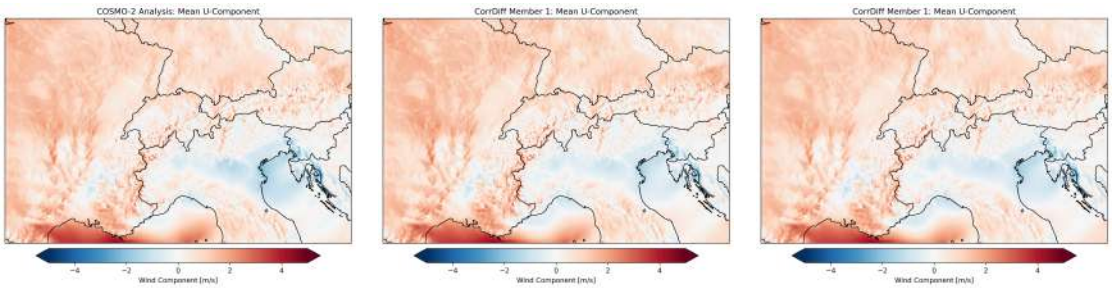


(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.12: Comparison of mean wind speed. This diagnostic evaluates whether broad spatial patterns of near-surface wind magnitude are preserved under compressed-input inference.

A.1.4 Wind Component Diagnostics

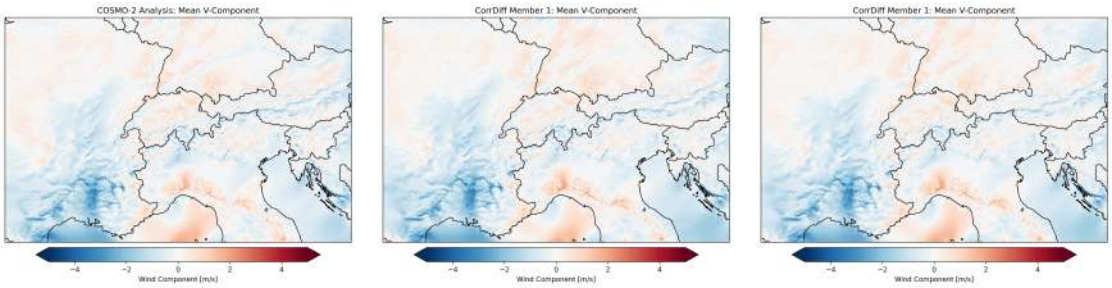
Mean U-Component



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.13: Comparison of mean zonal wind component. This component-wise diagnostic directly assesses whether the u-component structure is preserved under compressed-input CorrDiff inference.

Mean V-Component

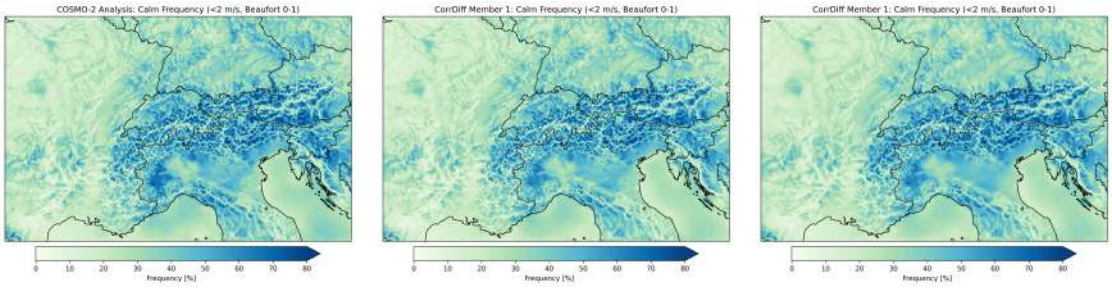


(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.14: Comparison of mean meridional wind component. This component-wise diagnostic directly assesses whether the v-component structure is preserved under compressed-input CorrDiff inference.

A.1.5 Wind Regime and Direction Diagnostics

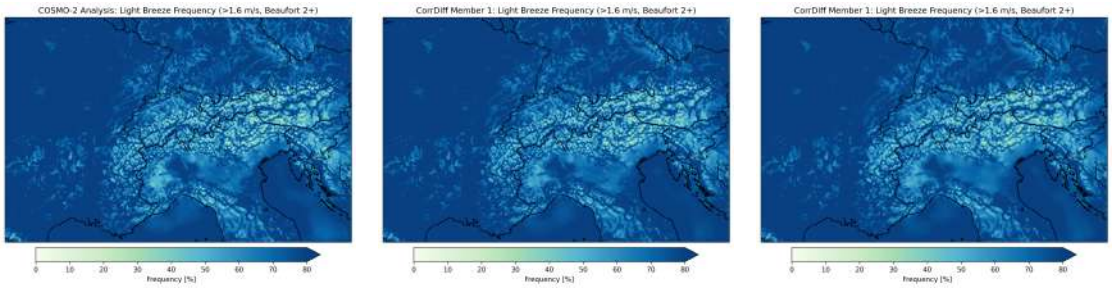
Calm Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.15: Comparison of calm frequency. This threshold-based wind diagnostic assesses whether weak-wind occurrence patterns are preserved under compressed-input CorrDiff inference.

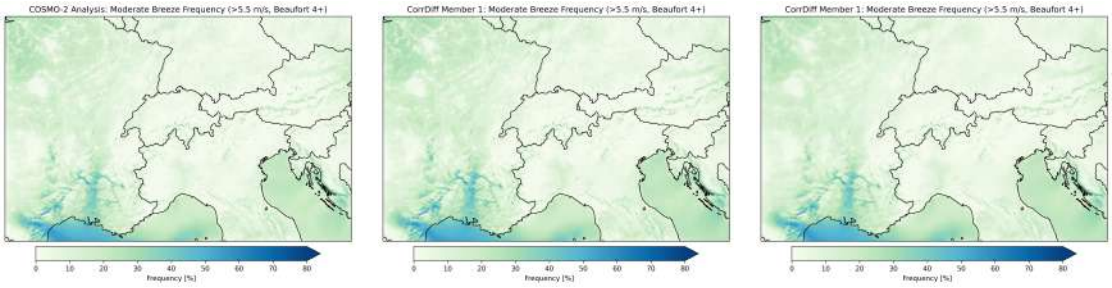
Light Breeze Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.16: Comparison of light breeze frequency. This diagnostic evaluates whether low but non-calm wind-regime patterns remain consistent between original-input and compressed-input CorrDiff predictions.

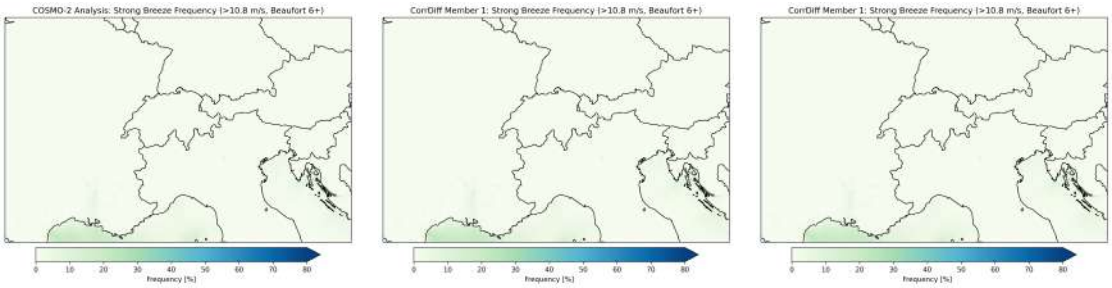
Moderate Breeze Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.17: Comparison of moderate breeze frequency. This diagnostic tests whether an intermediate wind-speed category is preserved after replacing original inputs with EBCC-compressed inputs.

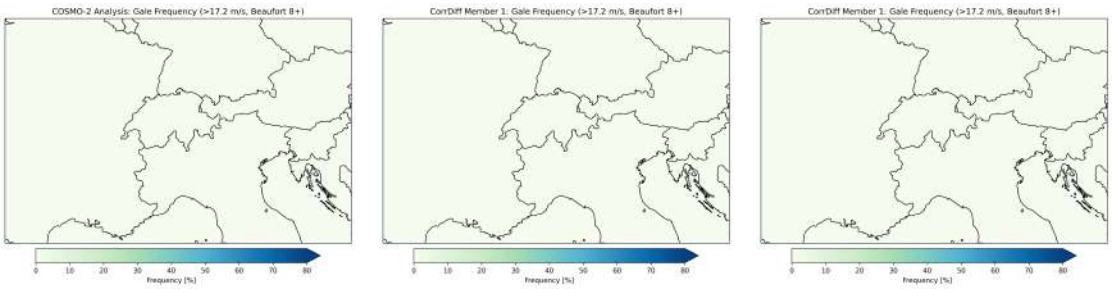
Strong Breeze Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.18: Comparison of strong breeze frequency. This diagnostic evaluates whether stronger near-surface wind-regime patterns remain consistent under compressed-input inference.

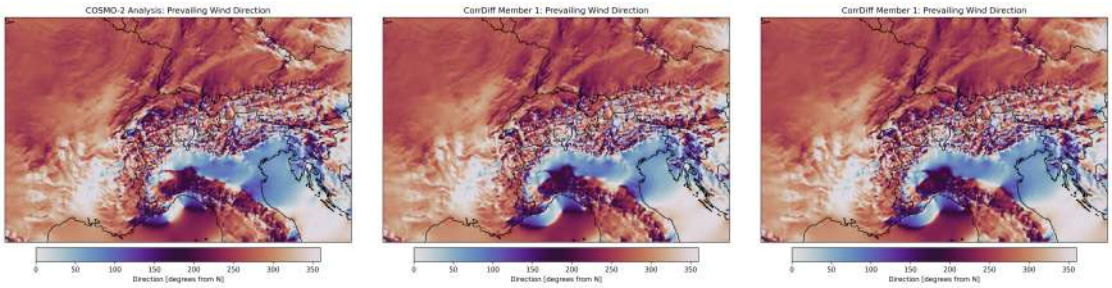
Gale Frequency



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.19: Comparison of gale frequency. This tail-oriented wind-regime diagnostic checks whether high-wind occurrence patterns are preserved by the compressed-input CorrDiff run.

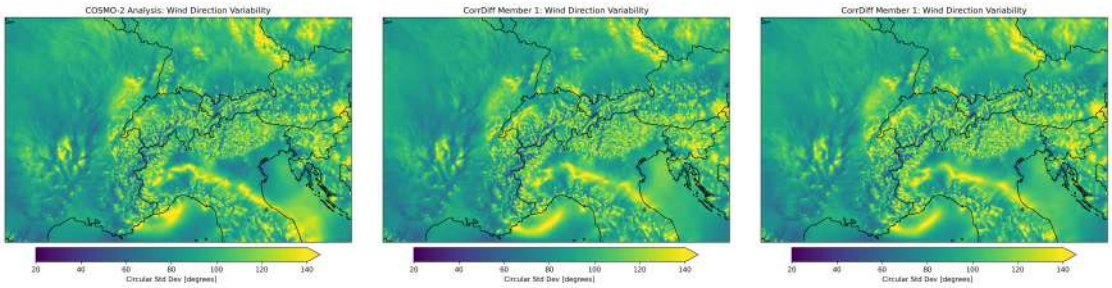
Prevailing Wind Direction



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.20: Comparison of prevailing wind direction. This directional diagnostic evaluates whether the dominant wind-direction structure is preserved after using EBCC-compressed inputs.

Wind Direction Variability

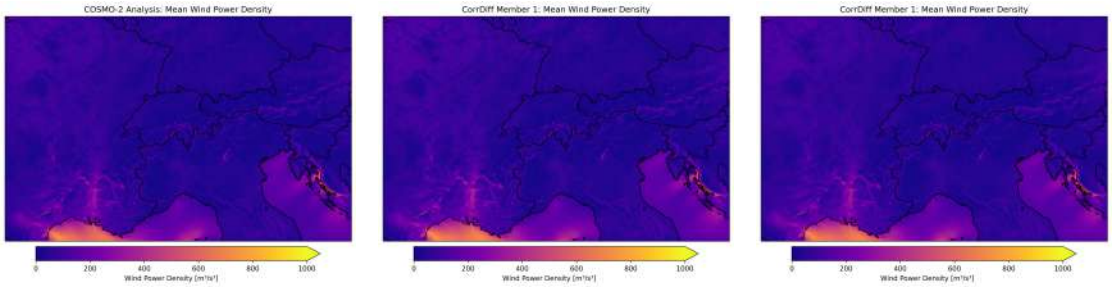


(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.21: Comparison of wind direction variability. This diagnostic assesses whether the spatial variability of wind direction remains consistent between original-input and compressed-input CorrDiff predictions.

A.1.6 Wind Energy Diagnostic

Mean Wind Power Density



(a) COSMO-2 target (b) Original input (c) Compressed input

Figure A.22: Comparison of mean wind power density. This nonlinear wind diagnostic provides an additional check that compressed wind inputs preserve downstream wind-energy-relevant spatial structure.

A.2 Diurnal-Cycle Diagnostics

A.2.1 2 m Temperature

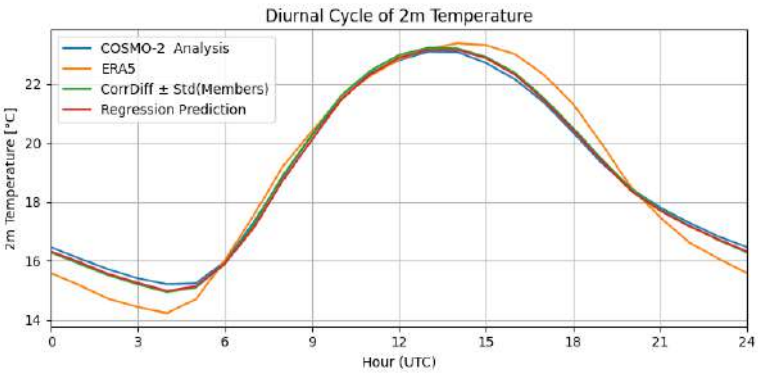


Figure A.23: Diurnal cycle of 2 m temperature from the CorrDiff evaluation. This diagnostic compares the daily near-surface temperature variability obtained from original and EBCC-compressed inputs.

A.2.2 Wind Speed

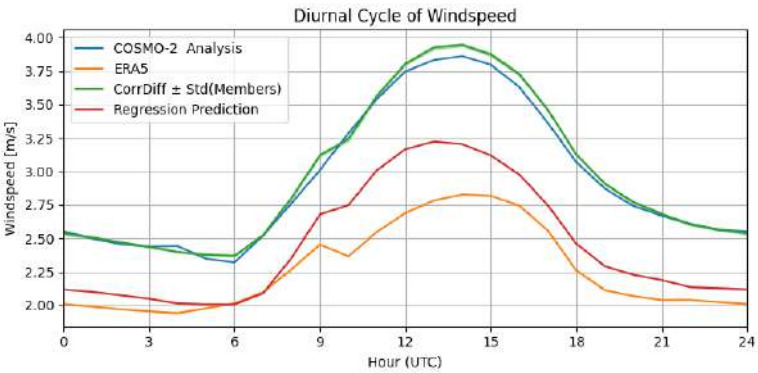


Figure A.24: Diurnal cycle of wind speed from the CorrDiff evaluation. This diagnostic assesses whether the daily phase and amplitude of wind-speed variability are preserved under compressed-input inference.

A.2.3 Precipitation Amount

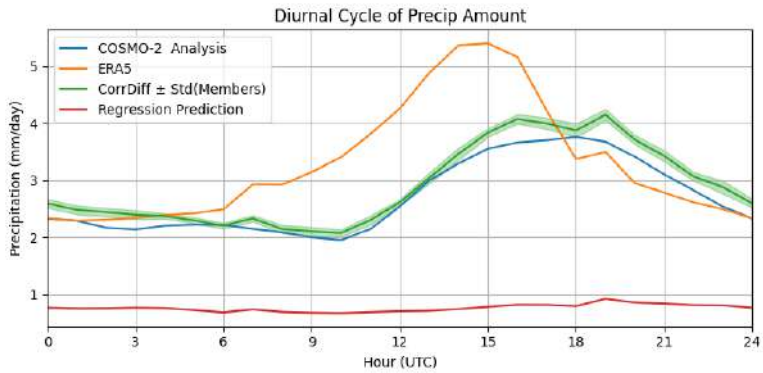


Figure A.25: Diurnal cycle of precipitation amount from the CorrDiff evaluation. This diagnostic tests whether the timing and magnitude of precipitation accumulation remain consistent between original-input and compressed-input runs.

A.2.4 Wet-Hour Frequency

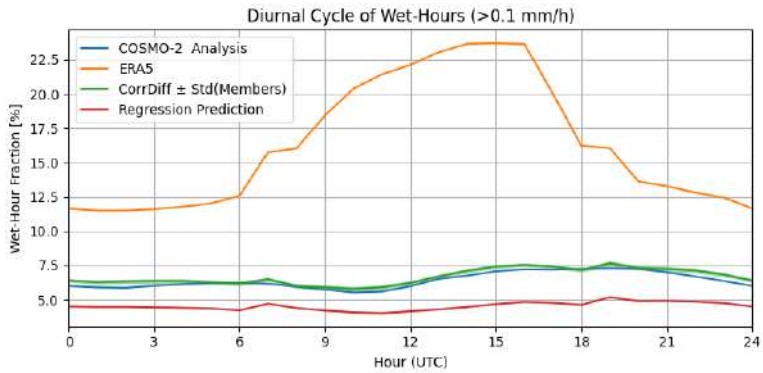


Figure A.26: Diurnal cycle of wet-hour frequency from the CorrDiff evaluation. Since wet-hour frequency is threshold-based, this diagnostic is sensitive to changes near the rain/no-rain boundary.

A.2.5 99th Percentile Precipitation

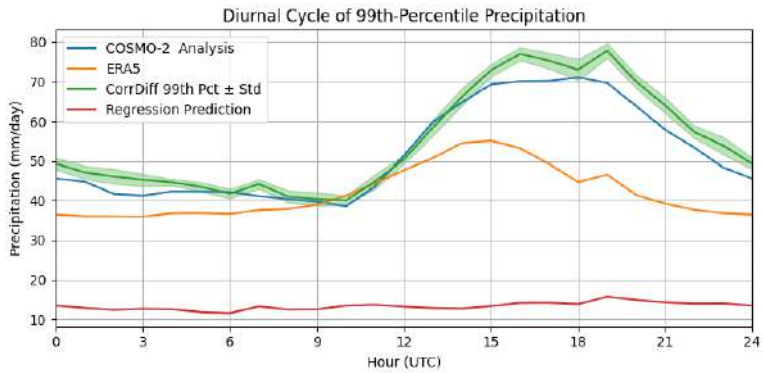


Figure A.27: Diurnal cycle of 99th percentile precipitation from the CorrDiff evaluation. This tail-oriented diagnostic tests whether high-percentile precipitation variability is preserved when using EBCC-compressed inputs.

A.3 Distributional and Tail Diagnostics

A.3.1 Domain-Mean Precipitation Over Land

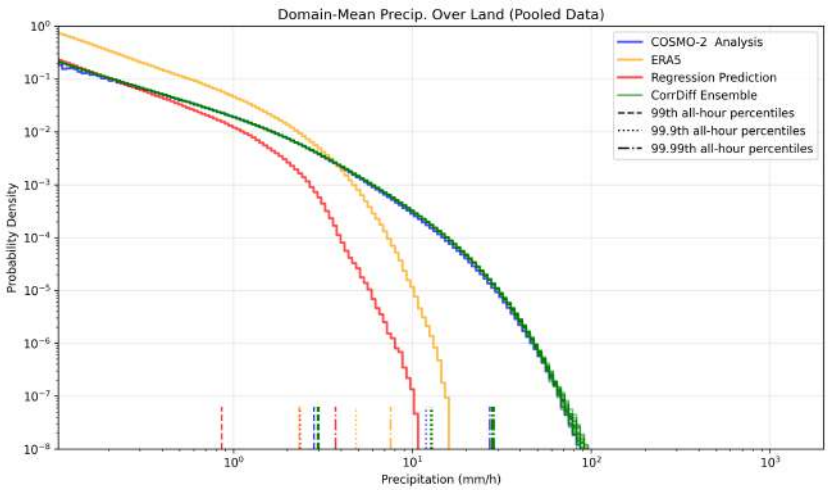


Figure A.28: Pooled precipitation distribution over land. This diagnostic compares the aggregate precipitation distribution and checks whether compressed-input CorrDiff predictions preserve precipitation statistics beyond selected maps.

A.3.2 Probability of Exceedance for Precipitation

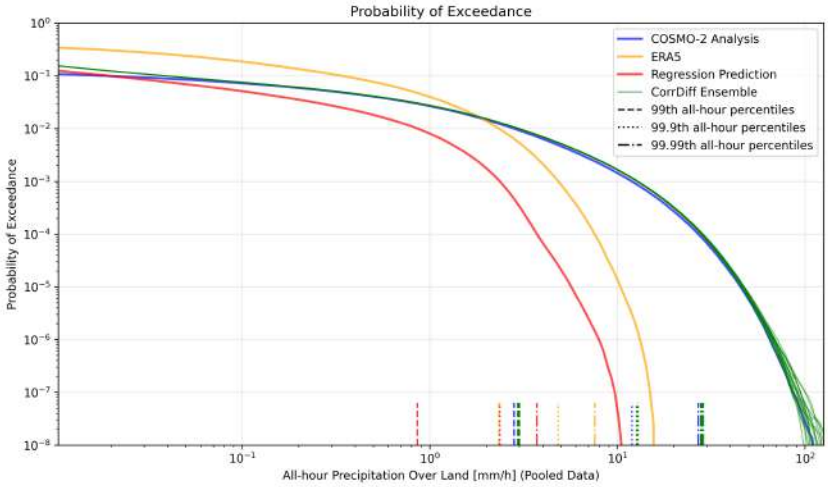


Figure A.29: Probability of exceedance for precipitation over land. This diagnostic directly compares the frequency with which precipitation values exceed a range of thresholds.

A.3.3 Probability of Exceedance for Wind Speed

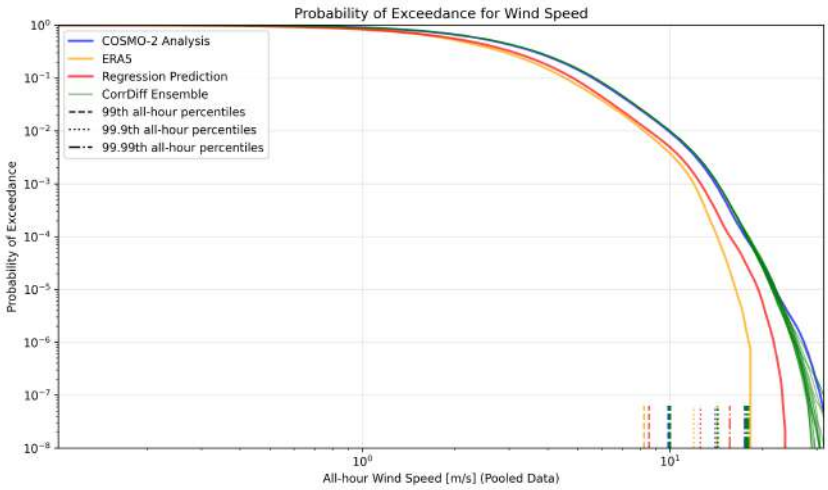


Figure A.30: Probability of exceedance for wind speed. This diagnostic assesses whether the tail behavior of derived wind speed is preserved under compressed-input inference.

A.3.4 Probability of Exceedance for $|10u|$

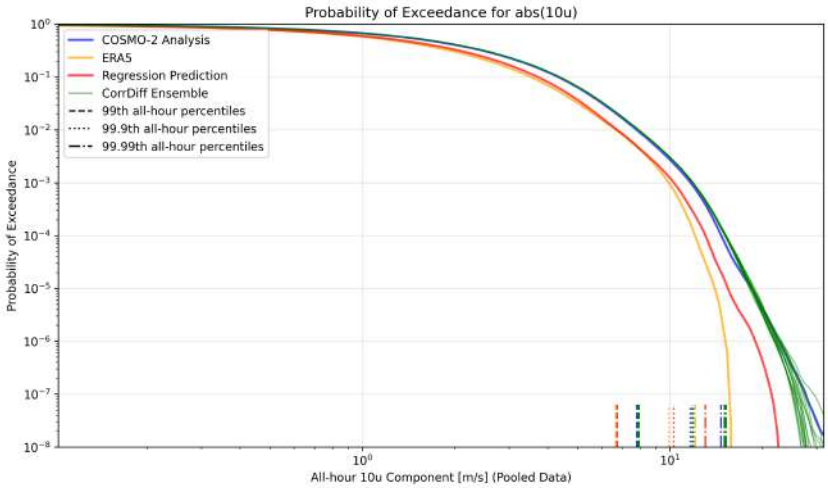


Figure A.31: Probability of exceedance for $|10u|$. This component-wise diagnostic checks whether the zonal wind contribution remains distributionally consistent after compression.

A.3.5 Probability of Exceedance for $|10v|$

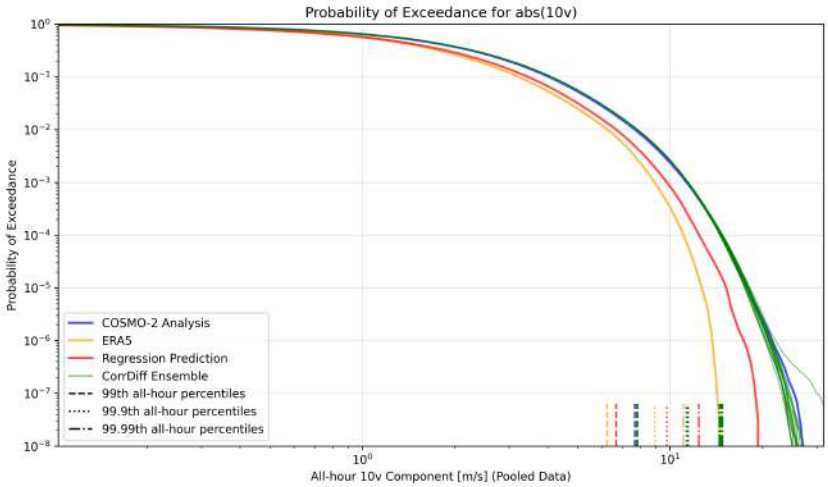


Figure A.32: Probability of exceedance for $|10v|$. This component-wise diagnostic checks whether the meridional wind contribution remains distributionally consistent after compression.

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[2] M. Mardani, N. Brenowitz, Y. Cohen, J. Pathak, C.-Y. Chen, C.-C. Liu, A. Vahdat, M. A. Nabian, T. Ge, A. Subramaniam, K. Kashinath, J. Kautz, and M. Pritchard, "Residual corrective diffusion modeling for km-scale atmospheric downscaling," *Communications Earth & Environment*, vol. 6, no. 1, p. 124, 2025.

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[4] B. M. Taye, "EBCC semester project: Error-bounded compression for machine learning applications in high resolution numerical weather prediction." `https://github.com/MeteoSwiss/EBCC-semester-project`, 2026. Project repository.

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